FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Acute Upper Respiratory Tract Infections

Tutorial | Week 1 · Breathing, fever & the clinical methods | Paediatrics | Dr Aliku

Most upper respiratory tract infections in children are viral and need only fluids, paracetamol and a clear nose; the skill is to spot the few that need an antibiotic (streptococcal tonsillitis, to prevent rheumatic fever; acute otitis media; bacterial sinusitis), the ones that need surgery or an ENT referral (quinsy, mastoiditis, chronic ear discharge), and the airway emergencies (croup, epiglottitis, diphtheria) where a calm child, oxygen, steroids or adrenaline, and the most skilled airway person save lives.

Must know before the ward round

1. Common cold (coryza): viral, self-limiting 7–10 days; no antibiotics; saline drops and suction, fluids, paracetamol. No cough syrups under 6 years (UCG 2023); no honey under 1 year.
2. Sore throat: most are viral (cough, runny nose, hoarseness, conjunctivitis). Exudate + tender anterior neck nodes + fever + no cough in a child ≥3 y suggests group A streptococcus (Centor/McIsaac, FeverPAIN).
3. Strep tonsillitis: phenoxymethylpenicillin 10–20 mg/kg 6-hourly (max 500 mg) for 10 days, or benzathine penicillin IM single dose (600,000 IU if <30 kg; 1.2 MU if ≥30 kg; UCG: 30,000 IU/kg if <30 kg). A full 10 days (or the injection) prevents rheumatic fever.
4. Acute otitis media: ear pain, fever, red bulging drum or pus <14 days. Amoxicillin 40–45 mg/kg twice daily for 5 days (WHO/IMCI; AAP high-dose 80–90 mg/kg/day); UCG 2023 lists 15 mg/kg 8-hourly. Dry the ear by wicking.
5. Chronic suppurative otitis media: pus ≥14 days. Dry by wicking 3 times daily + quinolone (ciprofloxacin) ear drops for 2 weeks; systemic antibiotics are not useful; test for HIV.
6. Mastoiditis: tender swelling behind the ear pushing the pinna forward and down → admit, ceftriaxone 50–80 mg/kg IV daily + metronidazole 7.5 mg/kg 8-hourly, urgent ENT referral.
7. Croup: barking cough, hoarse voice, inspiratory stridor, 6 months–3 years, parainfluenza (and measles in Uganda). Dexamethasone 0.15–0.6 mg/kg orally once (UCG: 0.15 mg/kg; 0.6 mg/kg max 12 mg if severe) or prednisolone 1–2 mg/kg.
8. Severe croup: nebulised adrenaline 1:1000, 0.4–0.5 mL/kg (max 5 mL) undiluted or with saline, repeat after 30 min if needed; observe at least 2–4 h for rebound; oxygen; call the airway expert.
9. Epiglottitis: toxic, drooling, tripod, muffled voice, little cough. Do NOT examine the throat, lie the child down or give IM injections; secure the airway in theatre; ceftriaxone 50–100 mg/kg IV once daily for 7–10 days. Hib vaccine (pentavalent) has made it rare.
10. Diphtheria: grey adherent pseudomembrane that bleeds when removed, bull neck, low fever, in an unimmunised child. Give diphtheria antitoxin early (20,000–100,000 IU by severity) plus penicillin or a macrolide for 14 days; isolate; prophylaxis and vaccination for contacts; watch for myocarditis.
11. Never examine the throat with a spatula in a child with stridor and drooling; never sedate a child with upper airway obstruction.
12. Antibiotics do not prevent pneumonia after a cold; over-prescription drives resistance (WHO AWaRe: amoxicillin and penicillin V are Access antibiotics).

1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology

5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup
9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

1. Definition

Upper respiratory tract infections (URTIs) are infections above the larynx (nose, sinuses, pharynx, tonsils, middle ear) plus the **laryngeal and tracheal** infections (croup, epiglottitis, bacterial tracheitis, diphtheria) that cause **stridor**. A Ugandan child has about **4–8 colds a year**, and URTIs are among the most common reasons for an outpatient visit, yet **almost all are viral**.

The clinician's job is to sort the child into one of four groups:

1. **Viral and self-limiting** (common cold, viral pharyngitis, most sore throats): reassure, symptom relief, safety-net advice, **no antibiotics**.
2. **Bacterial and treatable with a short oral course** (streptococcal tonsillitis, acute otitis media, bacterial sinusitis): amoxicillin or penicillin, given for the **right length of time**.
3. **Needs a procedure or specialist** (peritonsillar abscess, mastoiditis, chronic ear discharge, foreign body).
4. **Airway emergency** (severe croup, epiglottitis, diphtheria, bacterial tracheitis): keep the child calm, oxygen, steroid or adrenaline, and **the most skilled airway person**, now.

> CLINICAL PEARL
> The one thing never to forget: **a child with stridor and drooling must not be upset.** No spatula in the mouth, no lying flat, no IM injection, no separating from the mother. Agitation can turn a narrow airway into a closed one.

The syndromes

Syndrome	Site	Usual cause	Key feature
Common cold (coryza, rhinitis, rhinopharyngitis)	Nose and pharynx	Rhinovirus, RSV, adenovirus, coronavirus	Runny or blocked nose, sneezing, mild fever, 7–10 days
Pharyngitis / tonsillitis (pharyngotonsillitis)	Pharynx, tonsils	Viruses (most); **group A β-haemolytic streptococcus (GAS, *S. pyogenes*)** in 15–30% of school-age children	Sore throat, painful swallowing, ± exudate
Peritonsillar abscess (quinsy)	Between tonsil capsule and pharyngeal wall	GAS, anaerobes	Trismus, muffled "hot-potato" voice, uvula pushed to the opposite side
Acute otitis media (AOM)	Middle ear	*S. pneumoniae*, non-typeable *H. influenzae*, *M. catarrhalis*, viruses	Ear pain, fever, red **bulging** drum, or pus <14 days
Otitis media with effusion (glue ear)	Middle ear	After AOM, adenoids	Painless hearing loss, dull drum
Chronic suppurative otitis media (CSOM)	Middle ear with perforation	*Pseudomonas*, *S. aureus*, mixed	Ear discharge **≥14 days** (IMCI/UCG)
Mastoiditis	Mastoid air cells	Complication of AOM/CSOM	Tender swelling behind the ear, pinna pushed forward

Syndrome	Site	Usual cause	Key feature
Acute sinusitis	Paranasal sinuses	Viral; bacterial (pneumococcus, *H. influenzae*, *M. catarrhalis*)	Nasal discharge or cough **>10 days**, severe onset, or "double sickening"
Croup (laryngotracheobronchitis)	Larynx, trachea (subglottis)	**Parainfluenza**, RSV, influenza, **measles**	Barking cough, hoarseness, inspiratory stridor
Epiglottitis (supraglottitis)	Epiglottis, supraglottis	***H. influenzae* type b** (now rare after Hib vaccine), *S. pyogenes*, pneumococcus, *S. aureus*	Toxic, drooling, tripod posture, soft stridor, little cough
Bacterial tracheitis	Trachea	*S. aureus*, GAS	Croup that suddenly worsens, toxic, thick tracheal pus
Diphtheria	Tonsils, pharynx, larynx, nose, skin	Toxin-producing *Corynebacterium diphtheriae*	Grey adherent membrane, bull neck, myocarditis, neuropathy

Key terms

Term	Meaning
Stridor	Harsh, mainly **inspiratory** sound from narrowing of the extrathoracic airway (larynx, upper trachea)
Acute vs chronic ear discharge	**<14 days** = acute (AOM with perforation); **≥14 days** = chronic (CSOM)
Double sickening	Worsening after initial improvement of a cold: a clue to bacterial sinusitis
Pseudomembrane	Grey, adherent layer of dead epithelium, fibrin and bacteria that bleeds when removed: diphtheria

2. Epidemiology

- Children under five have **4–8 viral URTIs a year**; more in day care, crowded homes and households with smoke.
- **Antibiotic overuse**: in Ugandan outpatient surveys most children with a cough or cold leave with an antibiotic (often amoxicillin or cotrimoxazole) Verify *primary survey source*. Over-prescription drives resistance and side effects without benefit. **Amoxicillin and penicillin V are WHO AWaRe "Access" antibiotics**: when an antibiotic is needed, these are usually the right ones.
- **Rheumatic heart disease (RHD)**: Uganda has one of the highest burdens in the world; echocardiographic screening studies in Ugandan schoolchildren have found RHD in roughly **2–3 per 100** Verify *figure against the primary screening studies*. Each case began with a group A streptococcal throat infection that was not treated (see the rheumatic fever topic). Treating strep throat with a full course of penicillin is **primary prevention**.
- **Hib** vaccine has been in Uganda's pentavalent vaccine (DTP-HepB-Hib) since **2002**, and **PCV** since **2013**, given at **6, 10 and 14 weeks**: epiglottitis and pneumococcal ear disease have fallen sharply. An unvaccinated child with epiglottitis is a missed opportunity.
- **Diphtheria** still occurs in under-immunised communities (and in refugee settlements and border districts); DTP3 coverage in Uganda is high nationally but uneven by district.
- **Measles** causes croup, otitis media and pneumonia; Uganda gives **measles-rubella at 9 and 18 months** (UNEPI) and has ongoing outbreaks (MoH 2026).
- **CSOM** ("milk in the ear", as some families call it) is common, especially in HIV-positive and malnourished children; it is a leading cause of preventable hearing loss.

UGANDA CONTEXT

Current UNEPI schedule highlights relevant to URTIs: **BCG, OPV0 and hepatitis B birth dose** (introduced 2025) at birth; **DTP-HepB-Hib, PCV, OPV** at **6, 10, 14 weeks** (plus rotavirus and IPV); **measles-rubella at 9 and 18 months**; yellow fever at 9 months; **malaria vaccine at 6, 7, 8 and 18 months** in high-burden districts (since April 2025). Always check the child's card.

Services by level of care Verify *compiled from the source notes and local experience, not an MoH equipment list*

Level	Usually available	Often missing
HC III	Torch, spatula, paracetamol, amoxicillin, penicillin V, benzathine penicillin, malaria RDT, HIV test	Otoscope, pulse oximeter, oxygen, dexamethasone, nebuliser
HC IV / General Hospital	Otoscope, pulse oximeter, oxygen concentrator, nebuliser, dexamethasone, adrenaline, ceftriaxone, FBC, X-ray at hospitals	Throat and ear culture, ENT surgeon, anaesthetist skilled in difficult paediatric airways
Regional Referral Hospital (RRH)	Culture, ENT clinic, theatre, anaesthesia	CT at many RRHs; diphtheria antitoxin
National Referral Hospital (Mulago) and large PNFP hospitals (Mengo, Nsambya)	ENT surgery, CT, audiometry and tympanometry, bronchoscopy; diphtheria toxin testing at UNHLS/CPHL	Diphtheria antitoxin is scarce everywhere (request via MoH)

3. Aetiology and Risk Factors

Causes are listed by syndrome in section 1. In summary: the common cold, most sore throats and croup are **viral**; streptococcal tonsillitis, AOM, bacterial sinusitis, quinsy, mastoiditis, epiglottitis, bacterial tracheitis and diphtheria are **bacterial**.

Age patterns that narrow the cause

Age	Think of
Under 3 years with exudative tonsillitis	Usually **viral** (adenovirus, EBV); GAS uncommon
6 months–3 years with stridor	**Croup**
2–7 years (any age if unvaccinated) with toxic stridor	**Epiglottitis**
Under 5 years, fever, neck stiffness, drooling	**Retropharyngeal abscess**
3–15 years with exudative tonsillitis, no cough	**GAS**
Teenager with exudate, posterior nodes, splenomegaly	**Infectious mononucleosis (EBV)**
Toddler with unilateral foul nasal discharge	**Nasal foreign body**

Risk factors

Risk factor	Why it matters
Young age, immature immunity	More infections; short, horizontal Eustachian tube → AOM
Crowding, day care, sick siblings	More exposure
Indoor smoke (charcoal, firewood, tobacco)	Damages mucociliary clearance; more AOM
Bottle feeding lying flat	More AOM; **breastfeeding protects**
Large adenoids	Block the Eustachian tube → AOM, glue ear
Missed DTP-HepB-Hib, PCV or MR doses	**Diphtheria, epiglottitis, measles croup**, pneumococcal ear disease

Risk factor	Why it matters
HIV infection	Recurrent AOM, **CSOM**, oral candida
Malnutrition	CSOM, more severe infection
Previous rheumatic fever or RHD	Any GAS sore throat can cause recurrence: always treat
Traditional "uvulectomy" or tonsil cutting	Bleeding, infection

4. Pathophysiology

Why children get so many URTIs and ear infections

- Immature immunity, crowding and smoke exposure.
- **The Eustachian tube** in infants is **shorter, wider and more horizontal**, so nasopharyngeal secretions and bacteria reach the middle ear easily after a cold; large adenoids block it. That is why **AOM follows a cold** in a toddler.
- Bottle feeding lying flat and exposure to smoke increase AOM; breastfeeding protects.

Viral versus bacterial throat infections

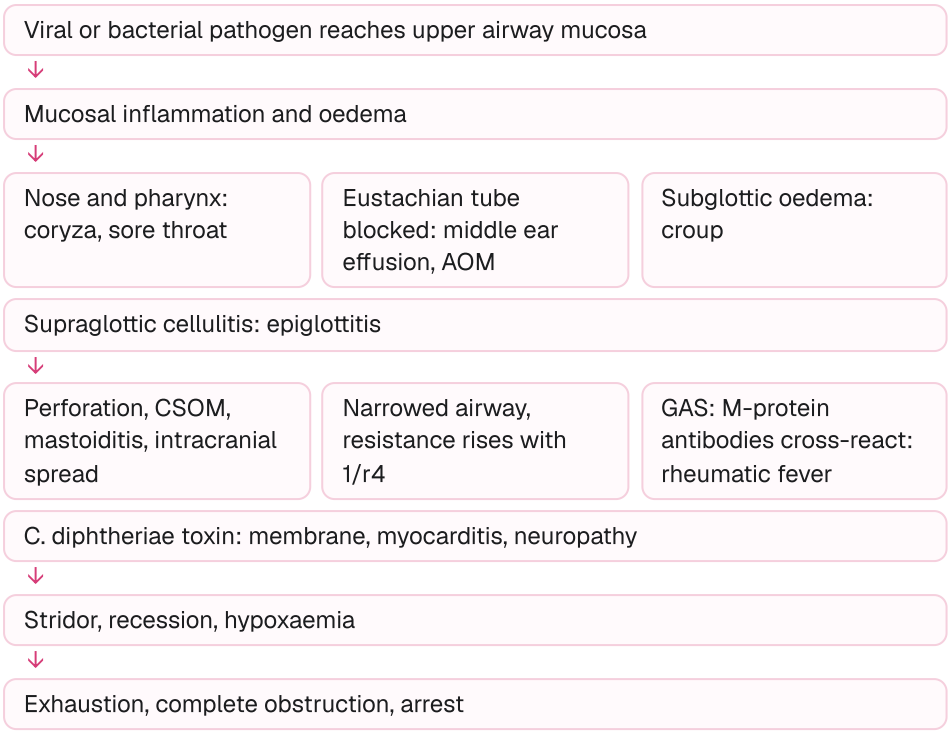
- Viruses inflame the whole upper airway: **cough, runny nose, hoarseness, conjunctivitis, ulcers**.
- **GAS** causes a localised exudative tonsillitis with **tender anterior cervical nodes, fever, and no cough**, in children **3–15 years** (rare under 3).
- **Molecular mimicry**: antibodies against streptococcal M protein cross-react with heart valves, joints and brain → **acute rheumatic fever** 2–4 weeks later. Treating within **9 days** of the sore throat prevents most ARF. Post-streptococcal **glomerulonephritis** is not prevented by antibiotics.

Why croup and epiglottitis obstruct a child's airway

- The **subglottis** (cricoid ring) is the narrowest part of a child's airway and is a complete ring of cartilage: mucosal swelling can only narrow the lumen.
- Resistance $\propto 1/\text{radius}^4$ (Poiseuille): **1 mm of oedema** in a 4 mm infant subglottis increases resistance about **16-fold**; in an adult, much less.
- Crying increases turbulent flow and negative pressure, which collapses the floppy extrathoracic airway further: **keep the child calm**.
- **Steroids** reduce mucosal oedema (onset 30 min–2 h, last 24–48 h). **Adrenaline** causes α-mediated vasoconstriction of the mucosa (onset 10 min, lasts about 2 h; symptoms can **rebound**).
- **Epiglottitis** is cellulitis of the supraglottis: the swollen epiglottis can fall back and **completely obstruct** the airway if the child lies down or gags.

Diphtheria

C. diphtheriae infected by a bacteriophage produces an **exotoxin** that kills epithelial cells (forming the **pseudomembrane** of dead cells, fibrin and bacteria) and enters the blood to damage the **heart (myocarditis, days 7–14)** and **nerves (palatal palsy, then cranial and peripheral neuropathy, weeks 2–8)**. **Antitoxin neutralises only unbound toxin**, so it must be given **early**, on clinical suspicion.



Feature links

- Mucosal inflammation of nose and pharynx → runny or blocked nose, sneezing, sore throat, cough.
- Inflammation of the whole upper airway (viral) → cough + coryza + hoarseness + conjunctivitis together.
- Localised GAS tonsillar infection → exudate, tender anterior cervical nodes, fever, no cough; palatal petechiae; toxin → scarlet fever (sandpaper rash).
- Cross-reacting anti-M-protein antibodies → acute rheumatic fever (carditis, arthritis, chorea) 2–4 weeks later.
- Blocked Eustachian tube → middle-ear pus under pressure → ear pain, fever, red bulging drum.
- Pressure → drum perforation → ear discharge; persistence ≥14 days → CSOM → conductive hearing loss, speech delay.
- Spread to mastoid air cells → post-auricular swelling pushing the pinna forward and down.
- Spread through bone or veins → meningitis, brain abscess, facial palsy, lateral sinus thrombosis.
- Sinus pus not draining → discharge or cough >10 days, facial pain; orbital spread → periorbital swelling, proptosis, painful eye movement.
- Subglottic oedema → narrowed lumen → inspiratory stridor; vocal cord inflammation → hoarse voice; tracheal inflammation → barking cough.
- Higher airway resistance → recession, tachypnoea, tachycardia; crying worsens collapse → stridor louder when upset.
- Supraglottic swelling → pain on swallowing → drooling, muffled "hot-potato" voice, tripod or sniffing posture; little cough because the larynx below is spared.
- Peritonsillar pus → trismus, uvula pushed to the opposite side.
- Diphtheria toxin killing epithelium → grey adherent pseudomembrane that bleeds when removed; neck oedema and nodes → bull neck.
- Circulating toxin → myocarditis (arrhythmia, heart block, heart failure, days 7–14) and neuropathy (palatal palsy with nasal regurgitation, then cranial, diaphragm and limb weakness, weeks 2–8).
- Severe obstruction → hypoxaemia → agitation, cyanosis; then exhaustion → softer stridor, falling consciousness.

5. Classification / Staging

IMCI classification for cough or cold (child 2 months–5 years)

Classification	Signs	Action
Severe pneumonia or very severe disease	Any general danger sign, or stridor in a calm child, or SpO_2 <90%	Pre-referral antibiotic, refer urgently
Pneumonia	Fast breathing and/or chest indrawing	Amoxicillin (see LRTI topic)
No pneumonia: cough or cold	None of the above	Home care, soothe the throat, **no antibiotics**; return if worse; follow up in 5 days if not improving; cough **>14 days** or recurrent wheeze → assess for TB, asthma, pertussis

IMCI ear problem box

Classification	Signs	Action
Mastoiditis	Tender swelling behind the ear	Refer
Acute ear infection	Pus <14 days or ear pain	Amoxicillin 5 days, wick the ear
Chronic ear infection	Pus ≥14 days	Wick, topical quinolone drops 2 weeks
No ear infection	None	No treatment

Croup severity (clinical, with the Westley score)

Grade	Features	Westley score
Mild	Barking cough, **no stridor at rest**, little or no recession, playing and feeding	≤2
Moderate	**Stridor at rest**, visible recession, not agitated	3–7
Severe	Stridor at rest with **marked recession**, agitation or lethargy, tachycardia	8–11
Impending respiratory failure	Cyanosis, **decreased consciousness**, stridor becoming softer or biphasic, poor air entry, exhaustion	≥12

Westley components: level of consciousness (0 or 5), cyanosis (0, 4 or 5), stridor (0–2), air entry (0–2), recession (0–3).

Sore throat scores (who to treat with an antibiotic)

Score	Items	Decision
McIsaac (modified Centor)	Fever >38 °C (+1); tonsillar exudate or swelling (+1); tender or swollen anterior cervical nodes (+1); **absence of cough** (+1); age 3–14 y (+1); age ≥45 (–1)	0–1: no antibiotic; 2–3: test if possible, otherwise clinical judgement; **4–5: treat**
FeverPAIN (NICE NG84)	**F**ever in past 24 h; **P**urulence; **A**ttend rapidly (≤3 days); **I**nflamed tonsils; **N**o cough or coryza	0–1 no antibiotic; 2–3 consider delayed; **4–5 antibiotic**

Bacterial sinusitis (AAP 2013; UCG): nasal discharge or daytime cough **>10 days** without improvement; **severe onset** (fever ≥39 °C with purulent discharge for ≥3 days); or **worsening after initial improvement** ("double sickening"). UCG: >1 week, unilateral facial pain, worsening.

Diphtheria antitoxin severity bands (WHO): pharyngeal or laryngeal disease <48 h; nasopharyngeal disease; extensive disease >3 days or bull neck (doses in section 9).

6. Clinical Features

History

Each question has a reason.

- **Main symptom and duration** → runny nose, sore throat, ear pain or discharge, noisy breathing, barking cough, drooling, voice change. Cold <10 days = viral; nasal symptoms >10 days = possible bacterial sinusitis; ear pus **≥14 days = CSOM**; cough **>14 days** = TB, asthma, pertussis.
- **Cough present?** → yes favours viral (and croup); absent favours GAS and epiglottitis.
- **Fever pattern** → high fever and toxicity (epiglottitis, tracheitis, quinsy, mastoiditis); low-grade (croup, diphtheria).
- **Onset** → hours (epiglottitis, foreign body, anaphylaxis); 1–2 days after a cold (croup); sudden choking (foreign body).
- **Eating and drinking** → refusal to swallow saliva = drooling (epiglottitis, quinsy, retropharyngeal abscess).
- **Drooling, voice change, trismus** → epiglottitis, quinsy, retropharyngeal abscess.
- **Immunisation card** → DTP-HepB-Hib ×3, PCV ×3, MR ×2 (UNEPI). Missed doses change the differential: unvaccinated → think diphtheria, epiglottitis, measles.
- **Contacts** → sick siblings, school outbreaks, measles in the village, diphtheria contacts, epiglottitis (Hib household prophylaxis).
- **Previous sore throats, joint pains, heart murmur** (ARF/RHD) → any sore throat must be treated; check secondary prophylaxis. Previous ear discharge.
- **HIV status, nutrition and growth (Z-scores), feeding** → recurrent and chronic ear disease; need for cotrimoxazole and ART.
- **Smoke exposure, day care** → recurrence risk and prevention advice.
- **Drugs given** → antibiotics already taken (how long?): failure of amoxicillin after 48–72 h in AOM → amoxicillin-clavulanate; herbal treatments; "uvulectomy" or tonsil cutting by traditional healers (bleeding, infection).

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: observe first, count breathing and listen to the chest while the child is quiet, and leave **distressing parts (ears and throat) to last**. **If there is stridor or drooling, do not examine the throat at all** and do not separate the child from the mother.

Introduction, consent and preparation

- **Introduction and identity**: introduce yourself; confirm the child's name and age and the carer's relationship.
- **Explanation and consent**: explain what you will do; consent from the parent and assent from the child.
- **Hand hygiene** before touching the child.
- **Position and exposure**: on the parent's lap, sitting up; never lie a child with stridor flat; expose the neck and chest in a warm room.

General inspection (end of the bed)

Sign	Mechanism	What it means
Well and playing vs toxic and still	Systemic bacterial infection	Toxic: epiglottitis, tracheitis, quinsy, mastoiditis
Tripod or sniffing posture, drooling	Supraglottic swelling, painful swallowing	**Epiglottitis**: leave the throat alone
Stridor at rest vs only when upset	Degree of subglottic narrowing	Stridor at rest = at least moderate croup

Sign	Mechanism	What it means
Recession (suprasternal, subcostal, intercostal)	High resistance on a compliant chest wall	Severity
Agitation, then drowsiness	Hypoxaemia, exhaustion	Severe; impending failure
Central cyanosis	Hypoxaemia	Impending respiratory failure
Voice: hoarse vs muffled	Vocal cords (croup) vs supraglottis (epiglottitis, quinsy)	Localises the lesion
Hydration, nutrition	Poor intake; chronic disease	Dehydration; malnutrition with CSOM

Vital signs and growth: temperature, RR, HR, **SpO_2** (correct probe, calm child), weight; plot weight-for-age and height-for-age Z-scores.

Hands, face and neck

Sign	Mechanism	What it means
Rash: sandpaper, flushed cheeks, circumoral pallor	GAS erythrogenic toxin	Scarlet fever
Rash, conjunctivitis, Koplik spots	Measles	Measles croup or otitis; vitamin A, isolate
Urticaria, lip or tongue swelling	Mast-cell degranulation	Anaphylaxis: IM adrenaline
Periorbital swelling, proptosis, painful eye movement	Spread from ethmoid sinus	Orbital cellulitis: admit
Unilateral foul nasal discharge	Retained object	Nasal foreign body
Tender anterior cervical nodes	Draining GAS tonsillitis	Supports GAS
Posterior and generalised nodes	Systemic EBV	Infectious mononucleosis
Bull neck (diffuse neck swelling)	Toxin-induced oedema and nodes	Diphtheria
Torticollis, neck stiffness	Retropharyngeal pus	Retropharyngeal abscess (or meningitis)
Facial palsy	Spread from the middle ear	Complicated otitis media

Respiratory system and chest

- **Inspection**: RR for a full minute, recession, chest indrawing.
- **Palpation and percussion**: usually normal in URTI; dullness → pneumonia.
- **Auscultation**: stridor transmitted to the chest is not wheeze; listen for crackles and bronchial breathing to **exclude pneumonia** (fast breathing, indrawing, crackles: see LRTI topic).

Completing the examination

- **Heart**: murmurs (RHD), gallop or arrhythmia (diphtheritic myocarditis).
- **Abdomen**: splenomegaly (mononucleosis).
- **Sinuses**: tenderness, periorbital swelling.
- **Ears, last**: otoscopy (drum colour, light reflex, bulging or retraction, perforation, discharge); mastoid swelling or tenderness behind the ear.
- **Throat, last, and only if there is no airway concern**: torch and spatula; tonsil size, exudate, **membrane** (grey, adherent, bleeds when lifted: do not pull it off), uvula position, petechiae on the palate, trismus.
- **Thank** the parent and child, **wash hands**, **summarise** with the syndrome and severity.

Document these

Positives	Negatives
Stridor at rest or not; recession; SpO_2	"No drooling, no toxicity; drinking well"
Tonsils: size, exudate; nodes: tender anterior	"No membrane; uvula central; no trismus"
Ear drum: colour, bulging, perforation, discharge duration	"No mastoid swelling or tenderness; no facial palsy"
Immunisation status	"No fast breathing or chest indrawing"

Red flags

RED FLAG

Airway danger: stridor at rest, drooling, tripod posture, toxic look, cyanosis or SpO_2 <92%, decreasing consciousness, a soft or disappearing stridor in a tiring child, a grey membrane in the throat, a bull neck. Call the most senior clinician and anaesthetist/ENT, keep the child with the mother, give oxygen without upsetting the child.

RED FLAG

Spread beyond the ear or sinus: swelling behind the ear, headache, neck stiffness, drowsiness or convulsions (meningitis, brain abscess), facial palsy, eye swelling with painful or reduced eye movement or proptosis (orbital cellulitis). Admit, IV antibiotics, refer.

- **General danger signs** (IMCI): unable to drink or breastfeed, vomits everything, convulsions, lethargic or unconscious → severe disease, refer.
- **Stridor in a calm child** is classified as very severe disease in IMCI.

Thinking like a clinician

- **"Cough + runny nose + sore throat = viral."** The more sites involved, the more likely viral.
- **"Barking cough = croup; drooling without cough = epiglottitis."** Croup children cough and want to drink; epiglottitis children sit still, drool and avoid swallowing.
- **"Croup that suddenly gets worse with high fever and toxicity"** = **bacterial tracheitis** (does not respond to adrenaline): needs airway and IV antibiotics.
- **Stridor in an unimmunised child with a grey membrane** = **diphtheria** until proven otherwise: give antitoxin on clinical suspicion; don't wait for culture.
- **Under 3 years with exudative tonsillitis** is usually viral (adenovirus, EBV); GAS is uncommon.
- **Exudative tonsillitis + generalised lymphadenopathy + splenomegaly** in a teenager = **infectious mononucleosis**; amoxicillin causes a rash.
- **Unilateral foul-smelling nasal discharge** in a toddler = **foreign body** in the nose.
- **Ear discharge ≥14 days** = CSOM: oral antibiotics don't help; drying and topical quinolone drops do. Test for HIV.
- **Fever in Uganda may be malaria, not the throat**: do an RDT in every febrile child.
- **"No antibiotics" is a treatment**, and it needs an explanation and a safety net: what to watch for and when to come back.
- **Common student mistakes**: reaching for the spatula in a drooling child; giving five days of penicillin for strep throat; giving oral antibiotics for chronic ear discharge; discharging 30 minutes after adrenaline; prescribing cough syrup to a toddler.

7. Differential Diagnoses

Of acute stridor

Differential	Distinguishing clinical feature	Distinguishing investigation
Croup	6 m–3 y, coryza 1–2 days, **barking cough**, hoarse, mild fever, not toxic, wants to drink	Clinical
Epiglottitis	2–7 y (any age if unvaccinated), **sudden high fever, toxic, drooling, tripod**, muffled voice, **little or no cough**	Clinical; airway in theatre
Bacterial tracheitis	Starts like croup then **sudden deterioration**, high fever, toxic, copious secretions, no response to adrenaline	Bronchoscopy (pus, membrane)
Diphtheria	Unimmunised, low fever, **grey adherent membrane**, bull neck, hoarseness, myocarditis later	Throat swab culture; ECG
Inhaled foreign body (laryngeal)	Sudden choking, no fever, any age (toddlers)	X-ray (if radiopaque), laryngoscopy
Retropharyngeal abscess	<5 y, fever, neck stiffness or torticollis, drooling, muffled voice	Lateral neck X-ray (widened prevertebral space), CT
Anaphylaxis / angioedema	Trigger, urticaria, lip or tongue swelling, wheeze, hypotension	Clinical
Measles croup	Rash, conjunctivitis, Koplik spots, unvaccinated	Clinical
Laryngomalacia (chronic)	Stridor from early weeks, worse lying supine or feeding, thriving	Laryngoscopy (ENT)

Of sore throat

Differential	Distinguishing clinical feature	Distinguishing investigation
Viral pharyngitis	Cough, coryza, conjunctivitis, hoarseness, ulcers	Clinical
GAS tonsillitis	3–15 y, fever, exudate, tender anterior nodes, no cough, palatal petechiae, scarlet rash	RADT or culture where available
Infectious mononucleosis (EBV)	Teenager, exudate, posterior nodes, **splenomegaly**, fatigue	FBC (atypical lymphocytes), monospot
Diphtheria	Grey membrane that bleeds when removed, bull neck, unimmunised	Culture
Peritonsillar abscess	Trismus, uvula deviated, hot-potato voice	Clinical; aspiration
Oral candidiasis / herpes gingivostomatitis	White plaques (HIV, antibiotics) / anterior mouth ulcers and gingivitis	Clinical; HIV test

Of ear pain or discharge

Differential	Distinguishing clinical feature	Distinguishing investigation
Acute otitis media	Pain, fever, red bulging drum; pus <14 days after perforation	Otoscopy
Otitis externa	Pain on moving the pinna, swollen canal, drum normal	Otoscopy
CSOM	Painless discharge ≥14 days, perforation	Otoscopy, swab
Mastoiditis	Post-auricular swelling, pinna pushed forward and down	Clinical; CT
Foreign body	Toddler, unilateral, visible object	Otoscopy

Differential	Distinguishing clinical feature	Distinguishing investigation
Referred pain	Tonsillitis, dental abscess, teething	Examine throat and teeth

8. Clinical Workup

Most URTIs need no laboratory test: the diagnosis is made by examining the throat, ears and neck. Never delay airway care for a test. "Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Bedside

Test	Why / how	Expected finding	How it changes management	Available at
Pulse oximetry	Severity in croup or epiglottitis; correct probe size, calm child	<92%: severe croup or airway compromise	Oxygen and senior review	HC IV and above; often missing at HC III
Croup score (Westley)	Consciousness, cyanosis, stridor, air entry, recession	Grades severity (section 5)	Steroid vs adrenaline vs airway team; repeat after treatment	All levels (clinical)
Centor/McIsaac or FeverPAIN score	Probability of GAS (section 5)	4–5 = high probability	Whether to treat with penicillin	All levels (clinical)
Otoscopy	Every child with ear pain, discharge or unexplained fever (see the respiratory/ENT examination topic)	Bulging drum (AOM) vs dull retracted drum (effusion) vs perforation (CSOM)	Amoxicillin vs wicking + drops vs ENT	HC IV and above; variable at HC III
Rapid antigen detection test (RADT) for GAS	Throat swab, 10 minutes	Positive	Treat; not widely available in Uganda	Private hospitals
Malaria RDT	Every febrile child in Uganda	Positive or negative	Fever may be malaria, not the throat	All levels
Random blood glucose	Sick or not feeding	Low	Treat hypoglycaemia	HC IV and above; variable at HC III

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Full blood count	Toxic child, mastoiditis, quinsy	Neutrophilia; atypical lymphocytes (mononucleosis)	Supports bacterial cause; not needed for simple URTI	HC IV and above
HIV test	CSOM, recurrent AOM, oral thrush, poor growth	Positive	ART, cotrimoxazole	All levels
ASOT	Suspected ARF or post-streptococcal GN (**not** for sore throat)	Raised	Evidence of recent GAS infection	RRH, NRH Verify

Imaging

Test	When	Expected finding	How it changes management	Available at
Lateral neck X-ray	**Only** if stable and diagnosis unclear; **never** in suspected epiglottitis with distress	"Thumb sign" (epiglottitis), "steeple sign" on AP (croup), widened prevertebral space (retropharyngeal abscess), radiopaque foreign body	Diagnostic clue; never delay airway care for a film; never send the child unaccompanied	Hospitals with X-ray
CT head, mastoid or sinus	Mastoiditis or sinusitis with intracranial or orbital complications	Abscess, sinus thrombosis	Surgery; neurosurgery	NRH and private hospitals

Specialised / referral-level

Test	When	Expected finding	How it changes management	Available at
Throat swab culture	Suspected GAS where available; outbreaks; **suspected diphtheria** (tell the lab: needs tellurite medium)	GAS; *C. diphtheriae* (toxin testing at the national reference lab, UNHLS/CPHL)	Confirms and triggers public health action; but **treat clinically** first	RRH, NRH
Ear swab culture	CSOM not responding, mastoiditis	*Pseudomonas*, *S. aureus*, *Proteus*	Guides topical or systemic choice	RRH, NRH
ECG	**Diphtheria** (daily in the first 2 weeks)	ST-T changes, heart block, arrhythmia	Myocarditis: bed rest, cardiac monitoring; predicts death	Hospitals Verify
Audiometry / tympanometry	Glue ear, CSOM, after mastoiditis	Conductive hearing loss	ENT referral, school support	Mulago and Mengo ENT clinics
Bronchoscopy / laryngoscopy	Suspected tracheitis, foreign body, laryngomalacia	Pus, membrane, object, floppy larynx	Removal, airway care, diagnosis	NRH

Putting it together

Model one-line summary

> "A 2-year-old boy weighing 11 kg (WAZ −0.8), fully immunised, HIV-exposed but negative at 18 months, with 2 days of coryza followed by a barking cough, hoarse voice and stridor at rest since midnight, alert, SpO_2 95%, moderate subcostal recession, no drooling and drinking well: **moderate croup**."

Problem list

1. Moderate viral croup (Westley 4).
2. Mild fever, drinking well.
3. Immunisations complete for age (check MR2 due at 18 months: given).

Working diagnosis: *acute laryngotracheobronchitis (croup), moderate severity.*

Model case presentation (ward round)

> "Sarah is a 6-year-old girl from Mukono weighing 19 kg (WAZ −0.6), fully immunised per UNEPI including pentavalent ×3, PCV ×3 and MR ×2, HIV-negative, on a family diet. She presents with 2 days of sore throat, painful swallowing and fever up to 39 °C, without cough or runny nose. No previous joint pains or known

> heart disease. On examination she is alert and drinking, T 38.8 °C, RR 22, HR 118, SpO_2 98%, with enlarged red tonsils with white exudate, tender anterior cervical nodes, uvula central, no membrane, no trismus; ears and chest normal; no murmur. McIsaac score 5. Assessment: **acute streptococcal tonsillitis**. Plan: benzathine penicillin 600,000 IU deep IM once (observed 30 minutes), paracetamol 280 mg (15 mg/kg) 6-hourly, fluids, counsel mother on the link with rheumatic fever and to return with joint pains, breathlessness or dark urine."

EXAM FAVOURITE

Worked example 1: scoring a sore throat. A 9-year-old girl: fever 38.6 °C, exudate on both tonsils, tender anterior cervical nodes, **no cough**. McIsaac score: fever 1 + exudate 1 + nodes 1 + no cough 1 + age 3–14 y 1 = **5** → high probability of GAS (about 50%). In Uganda, with a high RHD burden and no rapid test, **treat**: benzathine penicillin 600,000 IU IM once (weight 26 kg) or penicillin V for 10 days. A 4-year-old with runny nose, cough, hoarse voice and red throat scores 0–1 → viral, no antibiotic.

EXAM FAVOURITE

Worked example 2: the croup score after treatment. A 14-month-old (9 kg): stridor at rest (2), moderate recession (2), decreased air entry (1), agitated but no cyanosis (0) → **Westley 5, moderate**. Given dexamethasone 0.6 mg/kg (5.4 mg) orally. At 30 minutes: stridor at rest louder, marked recession (3), SpO_2 91% → score now 6–7 and worsening → **nebulised adrenaline 1:1000, 4.5 mL**, oxygen. At 20 minutes: no stridor at rest. **Must stay at least 2–4 hours** because adrenaline wears off; admit if stridor returns.

9. Management

Immediate / emergency

First 5 minutes

1. **ETAT triage**: is there **obstructed breathing (stridor at rest), severe respiratory distress, central cyanosis**? These are **emergency signs**. Move to the resuscitation area.
2. **Do not upset the child**: examine on the mother's lap; leave the throat alone if there is stridor or drooling.
3. **Oxygen** (blow-by or mask held by the mother) if SpO_2 <92% or severe distress.
4. **Decide the airway pathway**:
   - Barking cough + stridor → **croup** → steroid now; **nebulised adrenaline** if severe.
   - Toxic + drooling + tripod → **epiglottitis** → call anaesthetist/ENT, theatre, no interventions until the airway is secure.
   - Grey membrane, unimmunised → **diphtheria** → isolate, antitoxin, antibiotics, airway team.
   - Sudden choking → **foreign body**: if the child can cough, encourage coughing; if not effective, back blows and chest thrusts (infant) or abdominal thrusts (child >1 y).
   - Urticaria, angioedema, hypotension → **anaphylaxis** → IM adrenaline 0.01 mg/kg.
5. **If no airway emergency**: check for **general danger signs** (unable to drink or breastfeed, vomits everything, convulsions, lethargic or unconscious) and fast breathing or chest indrawing (pneumonia: see the LRTI topic).

ABCDE

- **A**: stridor? drooling? Keep the child upright on the parent's lap. No spatula. If the child obstructs before help arrives, **bag-valve-mask ventilation usually works**.
- **B**: SpO_2, RR, recession; oxygen held near the face without a struggle; nebulised adrenaline for severe croup.

- **C**: HR, capillary refill; IV access only once the airway is safe (not in suspected epiglottitis before the airway is secured).
- **D**: AVPU, glucose. **Never sedate** a child with upper airway obstruction.
- **E**: temperature, rash (measles, scarlet fever, urticaria), membrane, bull neck, mastoid swelling.

Definitive

A. Principles and antibiotic stewardship

1. **Most URTIs need no antibiotic.** Explain why (viral; antibiotics don't shorten it or prevent pneumonia; they cause diarrhoea, rash and resistance).
2. **Symptom relief**: paracetamol **15 mg/kg** 6-hourly (max 60 mg/kg/day) or ibuprofen 5–10 mg/kg 8-hourly (with food; avoid if dehydrated); fluids; saline nose drops; continue breastfeeding.
3. **Safety net**: return immediately if fast or difficult breathing, stridor at rest, unable to drink, very sleepy, convulsions, ear swelling, or not better in 5 days.
4. **When an antibiotic is needed, use the narrowest effective one** (penicillin V, amoxicillin) for the recommended time.

B. Common cold (UCG 2023 5.2.3)

Do	Don't
Continue breastfeeding; extra fluids; warm drinks	Antibiotics
Clear the nose with **saline drops** and gentle suction (bulb or cloth wick) before feeds	**Cough syrups, antihistamines, decongestant mixtures in <6 y** (UCG 2023: avoid; risk of toxicity)
Paracetamol for 2–3 days for fever or pain	Honey in **<1 y** (infant botulism)
Steam or honey (>1 y) as home remedies	Xylometazoline drops in infants; if used in older children (0.05%) **max 5 days** (rebound congestion)
Advise when to return (danger signs, fast breathing, ear pain)	

C. Pharyngitis and tonsillitis (UCG 2023 21.3.3–21.3.4)

- Decide using the McIsaac or FeverPAIN score (section 5).

> **UGANDA CONTEXT**
> In Uganda, with a very high burden of rheumatic heart disease and few rapid strep tests, **treat clinically suspected streptococcal tonsillitis** in children aged 3–15 years (exudate, tender anterior nodes, fever, no cough), and **always treat** a child who has had rheumatic fever or has RHD. UCG 2023 recommends a **full 10-day course** (or a single benzathine injection). NICE's 5-day option is for symptom relief in a low-RHD country and is **not** appropriate here.

- **First line**: **phenoxymethylpenicillin (penicillin V) 10–20 mg/kg 6-hourly for 10 days** (max 500 mg). Adherence falls after symptoms settle: stress the full course.
- **When adherence is doubtful**: **benzathine benzylpenicillin deep IM once** (600,000 IU if <30 kg; 1.2 MU if ≥30 kg).
- **Alternative**: **amoxicillin 50 mg/kg once daily (max 1 g) for 10 days**: equally effective, tastes better; avoid if mononucleosis is possible (rash).
- **Penicillin allergy**: erythromycin 12.5 mg/kg 6-hourly for 10 days (UCG 2023), or azithromycin 12 mg/kg once daily for 5 days (GAS macrolide resistance exists).
- Doses, routes and cautions: drug table.

Peritonsillar abscess (quinsy): admit; IV benzylpenicillin 50,000 IU/kg 6-hourly (or ceftriaxone) **plus metronidazole 7.5 mg/kg 8-hourly**; analgesia; IV fluids if unable to swallow; **needle aspiration or incision and drainage by ENT**; then oral antibiotics to 10–14 days. Tonsillectomy after recurrent quinsy.

Recurrent tonsillitis: tonsillectomy is considered for ≥7 episodes in 1 year, ≥5/year for 2 years or ≥3/year for 3 years (Paradise criteria), sleep-disordered breathing, or recurrent quinsy (ENT referral).

D. Ear infections

Acute otitis media

1. **Analgesia for everyone**: paracetamol or ibuprofen for 2–3 days.
2. **Antibiotic** (WHO Pocket Book/IMCI: all children with AOM in our setting): **amoxicillin**.
3. **Dry the ear by wicking** if there is pus: twist a clean absorbent cloth or tissue into a wick, insert, remove when wet, replace; repeat until dry; 3 times a day. No cotton buds, no oils, no herbal drops.
4. **Review after 5 days**: pain and fever settled? Pus persists → continue wicking; recurrent → check HIV, hearing.
5. **Failure after 48–72 h**: amoxicillin-clavulanate (amoxicillin component 45 mg/kg twice daily) or ceftriaxone 50 mg/kg IM daily for 3 days.

> UGANDA CONTEXT
> Amoxicillin doses for AOM differ: **WHO Pocket Book/IMCI: 40 mg/kg twice daily for 5 days** (the same dispersible tablets as pneumonia); **UCG 2023: 15 mg/kg 8-hourly for 5 days** (≈45 mg/kg/day); **AAP 2013: high-dose 80–90 mg/kg/day** in 2 doses (10 days if <2 y) because of pneumococcal resistance. In Ugandan wards and OPDs, the **IMCI 40 mg/kg twice-daily dose** is what most clinicians use. Quote UCG in the exam and know that higher doses exist.

Chronic suppurative otitis media (pus ≥14 days)

- **Dry mopping (wicking) 3 times daily** until the ear is dry (may take weeks).
- **Topical quinolone drops** (ciprofloxacin ear drops) 2–4 drops after each wicking, for **2 weeks** (IMCI/WHO; UCG 2023).
- UCG 2023 also describes gentle aural irrigation with dilute hydrogen peroxide (1 spoon in half a glass of clean lukewarm water) before wicking.
- **Systemic antibiotics are not recommended** (UCG 2023) unless complications.
- **Keep water out** of the ear when bathing or swimming. **HIV test.** Hearing assessment and ENT referral if persistent, cholesteatoma (white debris, foul smell), or polyps.

Mastoiditis (UCG 2023 21.1.6): admit urgently; **ceftriaxone 50–80 mg/kg IV/IM once daily for 10–14 days** (max 2–4 g) **plus metronidazole 7.5 mg/kg 8-hourly**; analgesia; urgent ENT for incision and drainage or mastoidectomy; watch for meningitis, brain abscess, facial palsy, sigmoid sinus thrombosis.

Glue ear (otitis media with effusion): most resolves in 3 months; hearing check; ENT referral if persistent >3 months with hearing loss or speech delay (grommets ± adenoidectomy).

> UGANDA CONTEXT
> UCG 2023 lists chlorphenamine and decongestant drops for glue ear; international guidelines do not recommend antihistamines or decongestants for glue ear.

E. Acute sinusitis (UCG 2023 21.2.4)

- **Diagnose bacterial sinusitis only if** the criteria in section 5 are met (>10 days, severe onset, or double sickening).

- **Treatment**: saline irrigation, analgesia; **amoxicillin** 15 mg/kg 8-hourly (UCG) or 40–45 mg/kg twice daily (high-dose), for **7–10 days**; amoxicillin-clavulanate if no improvement in 72 h. Look for a **dental focus** or **nasal foreign body**.
- **Orbital complications** (periorbital swelling, pain on eye movement, proptosis, reduced vision): admit; **IV ceftriaxone 50–80 mg/kg daily + cloxacillin 50 mg/kg 6-hourly** (or + metronidazole); urgent ophthalmology/ENT and CT.

F. Croup (UCG 2023 5.2.7)

1. **Keep the child calm on the parent's lap**; no throat examination; no unnecessary procedures.
2. **Steroid for every child with croup who has stridor** (and for barking cough in young infants or those far from care):
   - **Dexamethasone 0.15 mg/kg orally once** (UCG 2023 mild croup; RCH mild–moderate), or **0.6 mg/kg (max 12 mg) orally/IM/IV** for severe croup (RCH; Medscape: trials used 0.6 mg/kg; 0.15 mg/kg is effective).
   - Or **prednisolone 1–2 mg/kg orally once** (UCG 2023; may need repeating).
3. **Severe croup** (stridor at rest with marked recession, agitation, SpO_2 <92%):
   - **Nebulised adrenaline 1:1000: 0.4–0.5 mL/kg (max 5 mL)**, made up to 4 mL with saline if needed; repeat after 30 minutes if necessary (UCG 2023: 0.4 mg/kg, max 5 mg).
   - **Oxygen** (humidified, held near the face by the parent) if SpO_2 <92%.
   - IV/IM steroid if unable to take oral: dexamethasone 0.3–0.6 mg/kg IM (UCG: 300 µg/kg IM), or hydrocortisone (UCG 2023: <1 y 25 mg, 1–5 y 50 mg, 6–12 y 100 mg) slow IV/IM; repeat after 6 h if necessary.
   - **Observe for at least 2–4 hours after adrenaline** (rebound).
4. **Impending respiratory failure** (cyanosis, decreasing consciousness, exhaustion): 100% oxygen, repeated adrenaline, call the anaesthetist; **intubation in theatre with a tube 0.5–1 mm smaller than usual**; tracheostomy if intubation fails (UCG: nasotracheal intubation or tracheostomy).
5. **Admit** if: stridor at rest after treatment, needed adrenaline (observe ≥4 h, admit if stridor returns), age <6 months, SpO_2 <92%, poor drinking, lives far away or unreliable return.
6. **Discharge**: no stridor at rest, drinking, parents understand when to return (stridor at rest, recession, drooling, blue colour, drowsiness).
7. **No antibiotics** unless bacterial tracheitis suspected (then treat as epiglottitis). **No cough mixtures** (UCG). Humidified air ("steam") has **no proven benefit** and risks scalds: do not advise steam kettles.

> **UGANDA CONTEXT**
> UCG 2023 advises IV hydration with **half-strength Darrow's in 2.5% glucose** for severe croup. Most Kampala wards now use **standard maintenance fluids** (e.g. Ringer's lactate or 0.9% saline with 5% glucose for younger children) only if the child cannot drink. Many children with croup in Uganda have **measles**: check for rash, Koplik spots and vaccination status, give vitamin A and isolate.

G. Epiglottitis (UCG 2023 5.2.4)

1. **Recognise it**: toxic, high fever, drooling, tripod or sniffing posture, muffled voice, soft stridor, anxious, **little or no cough**.
2. **Do NOT**: look in the throat, lie the child flat, take blood, give IM injections, or send the child to X-ray unaccompanied. Keep with the mother; oxygen held near the face.
3. **Call** the most experienced anaesthetist and ENT surgeon; **secure the airway in theatre** (gas induction, child sitting; tracheostomy set open). Bag-valve-mask ventilation usually works if the child obstructs before help arrives.

4. **After the airway is secure**: IV line, blood culture, **ceftriaxone 50–100 mg/kg IV once daily for 7–10 days** (UCG 2023: 50 mg/kg; many centres use 80–100 mg/kg, max 2 g) ± cloxacillin if *S. aureus* suspected.
5. **Hib prophylaxis**: rifampicin 20 mg/kg once daily (max 600 mg) for 4 days to household contacts if there is an unvaccinated or incompletely vaccinated child <4 years, or an immunocompromised child, in the home (CDC/AAP).
6. **Check and complete immunisation** (pentavalent).

Bacterial tracheitis: treat as epiglottitis (airway in theatre, IV ceftriaxone ± cloxacillin); bronchoscopy to clear pus and membrane.

H. Diphtheria (UCG 2023 2.1.3; WHO)

1. **Isolate** (droplet and contact precautions) until 2–3 negative cultures (UCG: 3 negative swabs) or 48 h of appropriate antibiotics.
2. **Diphtheria antitoxin (DAT)** as soon as diphtheria is suspected (don't wait for culture), after a test dose per the product leaflet (equine serum: anaphylaxis possible; adrenaline ready). Dose by severity (WHO): **20,000–40,000 IU** for pharyngeal or laryngeal disease <48 h; **40,000–60,000 IU** for nasopharyngeal disease; **80,000–100,000 IU** for extensive disease >3 days or bull neck; given IV in 0.9% saline over 1 h (or IM). WHO Pocket Book 2013: **40,000 units IM/IV**. DAT is scarce: request through MoH and national stores.
3. **Antibiotic** for 14 days:
   - UCG 2023: **procaine benzylpenicillin 50,000 IU/kg IM once daily** until able to swallow, then **penicillin V** (<1 y 12.5 mg/kg; 1–6 y 125 mg; >6 y 250 mg) 6-hourly to complete **14 days**.
   - Penicillin allergy: **erythromycin 12.5 mg/kg 6-hourly** (50 mg/kg/day) for 14 days. WHO's 2024 clinical guidance suggests a **macrolide (azithromycin or erythromycin)** as first choice where available.
4. **Airway**: watch for laryngeal extension and membrane aspiration; tracheostomy may be needed. Avoid pulling off the membrane.
5. **Heart**: strict bed rest, **ECG** daily for 2 weeks, monitor for arrhythmia and heart failure (myocarditis typically days 7–14).
6. **Nerves**: watch for palatal palsy (nasal regurgitation), then cranial, respiratory and limb weakness (weeks 2–8): NG feeding, ventilatory support.
7. **Contacts**: throat swabs; **single dose benzathine penicillin IM** (UCG: 600,000 IU if <10 y, 1.2 MU if ≥10 y) or erythromycin for 7–10 days; check and complete DTP vaccination, booster if last dose >1 year ago; monitor for 7 days.
8. **Notify** the district surveillance focal person immediately (epidemic-prone disease). **Vaccinate the patient** after recovery (disease does not reliably confer immunity).

Drug table

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Paracetamol (pain, fever)	15 mg/kg, max 60 mg/kg/day; adult 1 g	Oral	6-hourly	2–3 days	Overdose risk if given with combination syrups
Ibuprofen (pain)	5–10 mg/kg; adult 400 mg Verify	Oral	8-hourly with food	2–3 days	Avoid if dehydrated or asthma with NSAID sensitivity
Phenoxymethylpenicillin (penicillin V) (GAS tonsillitis)	10–20 mg/kg per dose (UCG 2023), max 500 mg; or by age: 1–5 y 125 mg, 6–11 y 250 mg (NICE NG84, 6-hourly); adult 500 mg	Oral	**6-hourly** (or 12-hourly at double dose)	**10 days**	Adherence falls once symptoms settle: stress the full course

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Benzathine benzylpenicillin (GAS tonsillitis; diphtheria contacts)	GAS: **600,000 IU if <30 kg; 1,200,000 IU if ≥30 kg** (WHO/AHA; UCG 2023: 30,000 IU/kg if <30 kg, 1.2 MU adults). Diphtheria contacts (UCG): 600,000 IU <10 y, 1.2 MU ≥10 y	**Deep IM**	Once	Single dose	Best when adherence is doubtful; painful (can be mixed with lidocaine 1% per local protocol); have adrenaline ready; observe 30 min
Amoxicillin (AOM, sinusitis, GAS)	AOM: **40 mg/kg 12-hourly** (WHO/IMCI; 40–45 mg/kg), UCG 15 mg/kg 8-hourly, AAP 80–90 mg/kg/day; GAS: 50 mg/kg once daily or 25 mg/kg twice daily; max 1 g/dose; adult 500 mg–1 g Verify	Oral (dispersible)	12-hourly (8-hourly UCG; once daily for GAS)	5 days AOM (10 if <2 y on AAP); 7–10 days sinusitis; 10 days GAS	Rash if given in infectious mononucleosis
Amoxicillin-clavulanate (AOM or sinusitis failure)	45 mg/kg (amoxicillin component)	Oral	12-hourly	7–10 days	Diarrhoea
Erythromycin (penicillin allergy; diphtheria)	12.5 mg/kg, max 500 mg	Oral	6-hourly	10 days (GAS); 14 days (diphtheria); 7–10 days (contacts)	GI upset; UCG 2023 allergy alternative
Azithromycin (penicillin allergy)	12 mg/kg, max 500 mg	Oral	Once daily	**5 days**	GAS macrolide resistance exists
Benzylpenicillin (quinsy)	50,000 IU/kg	IV	6-hourly	Until swallowing, then oral to 10–14 days	With metronidazole; drainage by ENT
Procaine benzylpenicillin (diphtheria)	50,000 IU/kg, max 1.2 MU	IM	Once daily	Until can swallow (total 14 days with penicillin V)	Not IV
Ceftriaxone (epiglottitis, mastoiditis, orbital cellulitis, severe AOM failure)	50–100 mg/kg (epiglottitis); 50–80 mg/kg (mastoiditis, orbital); 50 mg/kg IM (AOM failure); max 2 g (4 g in meningitis)	IV/IM	Once daily	3 days (AOM failure); 7–10 days (epiglottitis); 10–14 days (mastoiditis)	In epiglottitis give only after the airway is secure
Cloxacillin (orbital cellulitis; suspected *S. aureus*)	50 mg/kg Verify *max*	IV	6-hourly	With ceftriaxone	Phlebitis
Metronidazole (mastoiditis, quinsy)	7.5 mg/kg, max 500 mg	IV/oral	8-hourly	10–14 days	Metallic taste, nausea

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Ciprofloxacin ear drops (CSOM)	2–4 drops	Topical	3 times daily after wicking	2 weeks	Topical quinolones are safe in children; dry the ear first
Dexamethasone (croup)	0.15 mg/kg (UCG, mild–moderate) to 0.6 mg/kg (severe), max 12 mg (RCH; 16 mg in some guides); IM 0.3–0.6 mg/kg (UCG 300 µg/kg)	Oral/IM/IV	Once (repeat after 12–24 h if needed)	1–2 doses	Oral as good as IM if swallowing
Prednisolone (croup)	1–2 mg/kg	Oral	Once	1 dose (repeat if needed)	–
Hydrocortisone (croup, cannot swallow)	UCG 2023: <1 y 25 mg, 1–5 y 50 mg, 6–12 y 100 mg	Slow IV/IM	Repeat after 6 h if needed	1–2 doses	–
Adrenaline 1:1000 (nebulised) (severe croup)	0.4–0.5 mL/kg, **max 5 mL** (UCG 2023: 0.4 mg/kg, max 5 mg)	Nebulised, made up to 4 mL with saline if needed	Repeat after 30 min if needed	–	Effect lasts about 2 h: **observe 2–4 h for rebound**; tachycardia
Adrenaline 1:1000 (IM) (anaphylaxis)	0.01 mL/kg (0.01 mg/kg), max 0.5 mg; adult 0.5 mg	IM thigh	Every 5 min if needed	–	First-line in anaphylaxis
Diphtheria antitoxin	20,000–100,000 IU by severity (WHO); WHO Pocket Book 40,000 units	IV over 1 h in 0.9% saline (or IM)	Once	–	Equine: test dose; adrenaline ready; scarce (MoH)
Rifampicin (Hib contacts)	20 mg/kg, max 600 mg	Oral	Once daily	4 days	Orange secretions; interactions
Xylometazoline 0.05% nasal drops	Older children only	Nasal	As per product	**Max 5 days**	Not in infants; rebound congestion

Adult doses where marked Verify *against UCG 2023 adult sections*; this topic is paediatric.

Referral criteria

Level	What to do	Refer up when
Home / VHT	Fluids, saline drops, paracetamol, safety-net advice	Fast or difficult breathing, stridor, unable to drink, ear swelling, not better in 5 days
HC III	Viral URTI care; penicillin V or benzathine for GAS; amoxicillin and wicking for AOM; wicking and quinolone drops for CSOM; pre-referral treatment	Any stridor at rest, drooling or toxicity, suspected diphtheria, mastoiditis, orbital swelling, quinsy, general danger sign → HC IV or hospital
HC IV / General Hospital	Dexamethasone, nebulised adrenaline, oxygen; admit croup; ceftriaxone + metronidazole for mastoiditis; IV antibiotics for quinsy	Epiglottitis, bacterial tracheitis or failing croup needing a skilled airway; mastoiditis or quinsy needing ENT drainage; diphtheria (antitoxin, isolation, ECG) → RRH
RRH	ENT, theatre, anaesthesia, culture, isolation	CT for intracranial or orbital complications; complex airway; tracheostomy care; audiology → NRH

Level	What to do	Refer up when
NRH (Mulago) and large PNFP hospitals	ENT surgery, CT, bronchoscopy, audiometry, PICU	n/a

Level split derived from the source notes and UCG levels of care Verify.

Referral checklist: steroid given (croup); oxygen during transfer; a health worker who can manage the airway accompanies the child; keep the child upright on the parent's lap; first antibiotic dose given (except in epiglottitis before the airway is safe); referral note with times and doses; phone ahead.

Monitoring and follow-up

- **Croup**: re-score (Westley) after steroid and 15–30 minutes after adrenaline; observe at least **2–4 hours** after adrenaline; SpO_2, RR, recession, conscious level.
- **Diphtheria**: **ECG daily for 2 weeks**; palatal and limb weakness for 8 weeks; airway.
- **AOM**: review at **5 days**; if pus persists, continue wicking; if recurrent, HIV test and hearing check.
- **CSOM**: review until dry; hearing assessment; ENT if not dry after 2 weeks of drops or cholesteatoma.
- **GAS tonsillitis**: counsel the parent to return with joint pains, breathlessness or dark urine (ARF, post-streptococcal GN).
- **Glue ear**: hearing check at 3 months.

10. Complications

Disease-related

Infection	Complications
Common cold	AOM, sinusitis, pneumonia, asthma exacerbation, conjunctivitis
GAS tonsillitis	**Acute rheumatic fever → RHD**, post-streptococcal glomerulonephritis, quinsy, retropharyngeal abscess, scarlet fever, invasive GAS
AOM / CSOM	Perforation, **hearing loss**, speech delay, **mastoiditis**, **meningitis**, brain abscess, facial palsy, lateral sinus thrombosis, cholesteatoma
Sinusitis	Periorbital and orbital cellulitis, cavernous sinus thrombosis, frontal bone osteomyelitis (Pott's puffy tumour), brain abscess
Croup	Hypoxia, respiratory failure, bacterial tracheitis, pneumonia
Epiglottitis	Complete airway obstruction, death, Hib meningitis, pneumonia
Diphtheria	**Airway obstruction**, **myocarditis** (heart block, heart failure, sudden death), **neuropathy** (palatal palsy, diaphragm palsy), renal failure; case fatality about **30%** in unvaccinated people without treatment (WHO)

Treatment-related

Complication	Cause	Recognition and action
Anaphylaxis	Benzathine penicillin; equine diphtheria antitoxin	Adrenaline ready; observe 30 min; DAT test dose
Rebound stridor	Nebulised adrenaline wears off after about 2 h	Observe 2–4 h; admit if stridor returns

Complication	Cause	Recognition and action
Rash	Amoxicillin in infectious mononucleosis	Avoid amoxicillin when EBV is possible
Diarrhoea	Amoxicillin-clavulanate, any antibiotic	Oral rehydration
Toxicity in young children	Cough syrups, antihistamines, decongestants	Avoid under 6 years (UCG 2023)
Rebound congestion	Xylometazoline >5 days	Limit to 5 days; not in infants
Infant botulism	Honey under 1 year	No honey under 1 year
Scalds	Steam kettles for croup	Do not advise steam
Complete obstruction	Throat examination, lying flat or upsetting a child with epiglottitis	Leave the throat alone; airway in theatre
Respiratory arrest	Sedation in upper airway obstruction	Never sedate
Antimicrobial resistance	Antibiotics for viral URTIs	Stewardship (section 9)

11. Prognosis

- **Common cold**: self-limiting in **7–10 days**; the cough can last **2–3 weeks**.
- **Viral pharyngitis and croup**: most recover within days; croup severe enough to need intubation is uncommon Verify *Ugandan data*.
- **GAS tonsillitis**: symptoms settle in about 3 days; the full 10-day course (or benzathine) prevents most rheumatic fever if started within **9 days**.
- **Glue ear**: most resolves within **3 months**.
- **CSOM**: drying may take weeks; untreated, it causes permanent conductive hearing loss and can lead to mastoiditis and intracranial infection.
- **Epiglottitis**: good if the airway is secured in time; death from complete obstruction if not.
- **Diphtheria**: case fatality about **30%** in unvaccinated people without treatment (WHO); myocarditis predicts death; survivors of neuropathy usually recover fully Verify.

12. Prevention

- **Vaccination (UNEPI)**: DTP-HepB-Hib and PCV at **6, 10, 14 weeks** (diphtheria, Hib epiglottitis, pneumococcal AOM and sinusitis); **measles-rubella at 9 and 18 months** (measles croup and otitis). Check every child's card and catch up.
- **Primary prevention of RHD**: treat streptococcal tonsillitis with a **full 10 days** of penicillin V or one benzathine injection.
- **Secondary prophylaxis** for children with previous ARF or RHD (see the rheumatic fever topic).
- **Breastfeeding**; avoid bottle feeding lying flat; reduce indoor smoke.
- **Contact prophylaxis**: rifampicin for Hib household contacts; benzathine or erythromycin, swabs and vaccination for diphtheria contacts; notify diphtheria.
- **Antibiotic stewardship**: no antibiotics for colds; amoxicillin and penicillin V (AWaRe Access) when one is needed.
- **Ear care**: keep water out of a perforated ear; no sticks, oils or herbal drops.

13. Special Populations

- **Infants**: no honey under 1 year; no cough syrups, antihistamines or decongestants under 6 years; no xylometazoline in infants; **croup under 6 months → admit**. Pull the pinna **down and back** for otoscopy.
- **Under 3 years with exudative tonsillitis**: usually viral; GAS uncommon.
- **Teenagers**: exudative tonsillitis with splenomegaly = mononucleosis; avoid amoxicillin.
- **HIV-positive children**: recurrent AOM, CSOM and oral candida; test every child with CSOM; continue cotrimoxazole and ART.
- **Malnourished children**: CSOM is common; more severe infection.
- **Previous ARF or RHD**: **always treat** a sore throat; check secondary prophylaxis.
- **Unimmunised children**: diphtheria, epiglottitis and measles move up the differential; vaccinate after recovery.
- **Measles**: croup and otitis; vitamin A, isolation.

14. Key Points

High-yield points

- Almost all URTIs are viral: fluids, paracetamol, saline drops, safety net, **no antibiotics**.
- Sort every child into one of four groups: viral, short oral antibiotic, procedure or specialist, airway emergency.
- **A child with stridor and drooling must not be upset**: no spatula, no lying flat, no IM injection.
- Strep throat in Uganda: **10 days** of penicillin V or one benzathine injection, to prevent rheumatic fever.
- AOM: amoxicillin 40 mg/kg twice daily for 5 days (IMCI) and wicking; CSOM (≥14 days): wicking and quinolone drops for 2 weeks, HIV test.
- Mastoiditis: admit, ceftriaxone + metronidazole, ENT.
- Croup: steroid for every child with stridor; nebulised adrenaline for severe croup; observe **2–4 h** afterwards.
- Epiglottitis: airway in theatre first, then ceftriaxone.
- Diphtheria: antitoxin on clinical suspicion, 14 days of antibiotic, isolation, ECG, contacts, notify.
- Check the immunisation card in every child.

Numbers & doses

Item	Value
Colds per year (under 5)	**4–8**
Bacterial sinusitis	Symptoms **>10 days**, severe onset ≥3 days, or double sickening
Acute vs chronic ear discharge	**<14 days** vs **≥14 days**
GAS age range	**3–15 years** (rare <3 y)
Penicillin V	**10–20 mg/kg 6-hourly × 10 days**
Benzathine penicillin	**600,000 IU (<30 kg); 1.2 MU (≥30 kg)** IM once (UCG: 30,000 IU/kg if <30 kg)
Amoxicillin (AOM)	**40 mg/kg 12-hourly × 5 days** (WHO/IMCI); UCG 15 mg/kg 8-hourly
Topical quinolone (CSOM)	After wicking, 3 times daily × **2 weeks**
Mastoiditis	Ceftriaxone **50–80 mg/kg** daily + metronidazole **7.5 mg/kg** 8-hourly × 10–14 days
Dexamethasone (croup)	**0.15–0.6 mg/kg** once (max 12 mg)
Prednisolone (croup)	**1–2 mg/kg** once

Item	Value
Nebulised adrenaline 1:1000	**0.4–0.5 mL/kg, max 5 mL**; observe **2–4 h**
Ceftriaxone (epiglottitis)	**50–100 mg/kg** IV daily × 7–10 days
Hib contact prophylaxis	Rifampicin **20 mg/kg** daily × **4 days** (max 600 mg)
DAT	**20,000–100,000 IU** by severity (WHO Pocket Book 40,000)
Diphtheria antibiotics	**14 days**
Diphtheria myocarditis	Days **7–14**; neuropathy weeks **2–8**
UNEPI	DTP-HepB-Hib + PCV at **6, 10, 14 weeks**; MR at **9 and 18 months**
Subglottic oedema	1 mm in a 4 mm airway → resistance about **16-fold**
Prevents ARF	Treat within **9 days** of the sore throat

Exam traps

- **"Amoxicillin for a cold to prevent pneumonia."** No: no benefit, only harm.
- **"Look at the throat to confirm epiglottitis."** Never: it can cause complete obstruction.
- **"Five days of penicillin for strep throat."** In Uganda: **10 days** (or benzathine) to prevent rheumatic fever.
- **"Oral antibiotics for chronic ear discharge."** No: wick and use topical quinolone drops.
- **"Steam inhalation for croup."** No proven benefit, risk of scalds.
- **"Send home 30 minutes after adrenaline because the stridor has gone."** No: observe **2–4 hours** for rebound.
- **"Wait for the culture before giving diphtheria antitoxin."** No: give on clinical suspicion.
- **"Cough syrup for a 2-year-old."** UCG 2023: avoid cough mixtures under 6 years.
- **"Antibiotics prevent post-streptococcal glomerulonephritis."** No: they prevent rheumatic fever, not GN.

Questions seniors ask

Viva questions

1. *How do you distinguish croup from epiglottitis?* Croup: 6 m–3 y, prodrome, barking cough, hoarse voice, mild fever, not toxic, drinking. Epiglottitis: sudden high fever, toxic, drooling, tripod, muffled voice, little cough, won't swallow.
2. *What is the dose of dexamethasone for croup?* 0.15–0.6 mg/kg orally once (UCG: 0.15 mg/kg; severe: 0.6 mg/kg, max 12 mg).
3. *Why must a child given nebulised adrenaline be observed for hours?* Its effect lasts about 2 hours, after which stridor can return (rebound) while the steroid is still taking effect.
4. *Why treat strep throat for 10 days?* To eradicate GAS from the pharynx and **prevent rheumatic fever**; symptoms settle in 3 days but eradication needs 10 days (or one benzathine injection).
5. *Define CSOM and give its treatment.* Ear discharge ≥14 days through a perforation; dry wicking 3 times daily + topical quinolone drops for 2 weeks; no systemic antibiotics; test HIV, assess hearing.
6. *What is the first drug in suspected diphtheria?* Diphtheria antitoxin, as early as possible on clinical suspicion, plus penicillin or erythromycin, isolation and contact prophylaxis.
7. *What vaccine prevents epiglottitis, and when is it given in Uganda?* Hib, in the pentavalent DTP-HepB-Hib at 6, 10 and 14 weeks.

On the ward round

- **This child has a red throat and a runny nose. Will you give an antibiotic?** No. Cough and coryza point to a viral pharyngitis; I'd give paracetamol, fluids and advice on when to return.

- **What features make you think of streptococcal tonsillitis?** Age 3–15, fever, tonsillar exudate, tender anterior cervical nodes, and no cough or coryza (McIsaac 4–5).
- **How do you treat moderate croup in an 18-month-old weighing 10 kg?** Keep calm on mum's lap; dexamethasone 1.5–6 mg (0.15–0.6 mg/kg) orally once; observe; nebulised adrenaline 4–5 mL of 1:1000 if it becomes severe; oxygen if SpO_2 <92%.
- **What would make you think this "croup" is something else?** Toxicity and high fever (epiglottitis, tracheitis), drooling (epiglottitis, retropharyngeal abscess), sudden onset without prodrome (foreign body), grey membrane (diphtheria), urticaria (anaphylaxis), or no response to adrenaline (tracheitis).
- **A 4-year-old has had ear discharge for a month. What do you do?** CSOM: teach wicking 3 times daily, ciprofloxacin ear drops for 2 weeks, keep water out, HIV test, check hearing, ENT referral if not dry or if cholesteatoma or complications.
- **What is the role of diphtheria antitoxin and why the hurry?** It neutralises circulating toxin before it binds to the heart and nerves; once bound it cannot be reversed, so give it on suspicion.
- **How does the pentavalent vaccine relate to this tutorial?** It contains diphtheria toxoid and Hib conjugate, preventing diphtheria and Hib epiglottitis (and Hib meningitis and pneumonia); PCV prevents much pneumococcal AOM, sinusitis and pneumonia.

Mnemonics

- **Croup "3 Bs and an S"**: **B**arking cough, **B**ad voice (hoarse), **B**etween 6 months and 3 years, **S**tridor.
- **Epiglottitis "4 Ds"**: **D**rooling, **D**ysphagia, **D**ysphonia (muffled voice), **D**istress (tripod), with toxicity.
- **FeverPAIN**: **F**ever, **P**urulence, **A**ttend rapidly, **I**nflamed tonsils, **N**o cough or coryza.
- **Centor "CENT"**: no **C**ough, **E**xudate, **N**odes (tender anterior), **T**emperature >38 °C (+ age for McIsaac).
- **Diphtheria "ABCDE"**: **A**ntitoxin, **B**ed rest (heart), **C**ulture and contacts, **D**rug (penicillin or erythromycin), **E**CG and **E**xclude airway obstruction.
- **Ear "<14 = acute, ≥14 = chronic"**.

OSCE tips

- **Otoscopy station**: choose the largest speculum that fits; **pull the pinna down and back in infants, up and back in older children**; hold the otoscope like a pen with your little finger resting on the child's head; describe the drum (colour, light reflex, bulging or retraction, perforation).
- **Throat station**: "I would not examine the throat in a child with stridor or drooling". Say it before you pick up the spatula.
- **Counselling station (viral URTI)**: explain viral cause, expected duration (cough up to 2–3 weeks), home care, why no antibiotic, and the danger signs to return for.
- **Ear-wicking demonstration**: twist a clean cloth or tissue into a wick, insert gently, remove when wet, replace with a clean wick, repeat until dry, 3 times daily; never leave a wick in or use sticks or oils.

15. References

1. Ministry of Health Uganda. Uganda Clinical Guidelines 2023 (5.2.3 Coryza; 5.2.4 Epiglottitis; 5.2.7 Croup; 21.1.4 Otitis media; 21.1.6 Mastoiditis; 21.2.4 Sinusitis; 21.3.3–21.3.5 Pharyngitis, tonsillitis, quinsy; 2.1.3 Diphtheria). Kampala: MoH; 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
2. World Health Organization. Pocket Book of Hospital Care for Children, 2nd edition. Geneva: WHO; 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
3. National Institute for Health and Care Excellence. Sore throat (acute): antimicrobial prescribing (NG84). 2018. https://www.nice.org.uk/guidance/ng84

4. The Royal Children's Hospital Melbourne. Croup (laryngotracheobronchitis): Clinical Practice Guideline. https://www.rch.org.au/clinicalguide/guideline_index/Croup_laryngotracheobronchitis/
5. Medscape (eMedicine). Croup. https://emedicine.medscape.com/article/962972-overview
6. Medscape (eMedicine). Epiglottitis. https://emedicine.medscape.com/article/963773-overview
7. Medscape (eMedicine). Pediatric Pharyngitis. https://emedicine.medscape.com/article/967384-overview
8. Medscape (eMedicine). Otitis Media. https://emedicine.medscape.com/article/994656-overview
9. World Health Organization. Diphtheria: fact sheet. https://www.who.int/news-room/fact-sheets/detail/diphtheria
10. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Management of the common cold in a 1-year-old (UCG 2023).	Continue breastfeeding, extra fluids, **saline nose drops** and gentle suction, paracetamol for 2–3 days, keep warm, return if danger signs. **No antibiotics, no cough syrups (<6 y), no honey (<1 y).**
2	IMCI classification 'No pneumonia: cough or cold' — what is it and what do you do?	Cough/cold **without** fast breathing, chest indrawing, stridor in a calm child or danger signs. Home care, soothe the throat, **no antibiotic**, return if worse, follow up in 5 days if not improving; cough >14 days → assess TB/asthma/pertussis.
3	McIsaac (modified Centor) score items.	Fever >38 °C (+1), tonsillar exudate/swelling (+1), tender anterior cervical nodes (+1), **absence of cough** (+1), age 3–14 y (+1), age ≥45 (–1). **4–5 = treat** for GAS.
4	FeverPAIN score.	**F**ever in past 24 h, **P**urulence, **A**ttend rapidly (≤3 days), **I**nflamed tonsils, **N**o cough or coryza. 0–1 no antibiotic; 4–5 antibiotic (NICE NG84).
5	First-line oral treatment of strep tonsillitis in a child (UCG 2023).	**Phenoxymethylpenicillin 10–20 mg/kg (max 500 mg) every 6 h for 10 days.** Penicillin allergy: erythromycin 12.5 mg/kg 6-hourly × 10 days (or azithromycin 12 mg/kg daily × 5 days).
6	Benzathine penicillin dose for strep tonsillitis.	**600,000 IU deep IM once if <30 kg; 1.2 MU if ≥30 kg** (WHO/AHA). UCG 2023: child <30 kg 30,000 IU/kg. Single dose; have adrenaline ready.
7	Why give a full 10-day course for GAS pharyngitis in Uganda?	To **eradicate GAS** and **prevent acute rheumatic fever/RHD** (treatment within 9 days of onset prevents most ARF). Uganda has a very high RHD burden. Post-streptococcal GN is not prevented.
8	Signs of a peritonsillar abscess (quinsy).	Severe unilateral throat pain, **trismus**, drooling, **muffled 'hot-potato' voice**, uvula pushed to the **opposite** side, fever, ear pain. Needs IV penicillin + metronidazole and **drainage** by ENT.
9	Define acute otitis media and give WHO/IMCI treatment.	Ear pain, fever, red **bulging** drum, or pus <14 days. **Amoxicillin 40 mg/kg twice daily × 5 days**, analgesia, dry by wicking, review in 5 days. (UCG 2023: 15 mg/kg 8-hourly × 5 days; AAP high-dose 80–90 mg/kg/day.)
10	Define CSOM and give its treatment.	Ear discharge **≥14 days** through a perforation. **Dry wicking 3×/day + topical quinolone (ciprofloxacin) drops for 2 weeks**; no systemic antibiotics; keep water out; HIV test; hearing check; ENT if persists.
11	How do you teach ear wicking?	Twist clean absorbent cloth/tissue into a wick, insert gently, remove when wet, replace with a clean wick, repeat until dry; **3 times daily**. Never cotton buds, sticks, oils or herbal drops.
12	Mastoiditis: signs and management (UCG 2023).	Tender swelling behind the ear, **pinna pushed forward and down**, fever, ear discharge. Admit; **ceftriaxone 50–80 mg/kg IV once daily + metronidazole 7.5 mg/kg 8-hourly × 10–14 days**; urgent ENT for drainage; watch for meningitis/abscess.
13	When do you diagnose bacterial sinusitis in a child?	Nasal discharge/daytime cough **>10 days** without improvement; **severe onset** (fever ≥39 °C + purulent discharge ≥3 days); or **double sickening**. Amoxicillin 7–10 days. Rare <5 y (UCG).

#	Question	Answer
14	Croup: typical age, cause and triad.	**6 months–3 years**; **parainfluenza** (also RSV, influenza, **measles**). **Barking cough, hoarse voice, inspiratory stridor**, after 1–2 days of coryza; worse at night.
15	Steroid doses for croup.	**Dexamethasone 0.15 mg/kg orally once** (UCG mild); **0.6 mg/kg (max 12 mg)** PO/IM/IV for severe (RCH). Or **prednisolone 1–2 mg/kg** once. Effect within 30 min–2 h.
16	Nebulised adrenaline for severe croup: dose and caution.	**Adrenaline 1:1000, 0.4–0.5 mL/kg (max 5 mL)** nebulised; repeat after 30 min if needed (UCG: 0.4 mg/kg max 5 mg). Effect lasts ~2 h: **observe at least 2–4 h** for rebound.
17	Westley croup score: components and cut-offs.	Consciousness (0/5), cyanosis (0/4/5), stridor (0–2), air entry (0–2), recession (0–3). **≤2 mild; 3–7 moderate; 8–11 severe; ≥12 impending respiratory failure.**
18	Epiglottitis: the classic picture.	Sudden **high fever**, toxic, **drooling**, **tripod/sniffing posture**, muffled voice, soft stridor, **little or no cough**, refuses to swallow. Classically **Hib** (rare since pentavalent vaccine).
19	Three things you must NOT do in suspected epiglottitis.	**Do not examine the throat** with a spatula, **do not lie the child down**, **do not give IM injections/take blood** before the airway is secured (UCG 2023). Keep with mother; airway in theatre.
20	Antibiotic for epiglottitis (UCG 2023).	**Ceftriaxone 50 mg/kg IV once daily for 7–10 days** (many centres 80–100 mg/kg, max 2 g), after the airway is secured. Rifampicin prophylaxis (20 mg/kg × 4 days) for household contacts if an unvaccinated child <4 y lives there.
21	Bacterial tracheitis: how does it present?	Starts like croup, then **sudden deterioration** with **high fever, toxicity, copious purulent secretions**, poor response to adrenaline. *S. aureus*/GAS. Airway (intubation) + IV antibiotics (e.g. ceftriaxone + cloxacillin).
22	Clinical features of diphtheria.	Unimmunised child, low fever, sore throat, **grey adherent pseudomembrane** that bleeds when removed, **bull neck**, hoarseness/stridor. Toxin → **myocarditis (days 7–14)** and **neuropathy** (palatal palsy, weeks 2–8).
23	Diphtheria treatment.	**Isolate**; **diphtheria antitoxin** on clinical suspicion (20,000–100,000 IU by severity; WHO pocket book 40,000); **procaine penicillin 50,000 IU/kg IM daily** then penicillin V (or erythromycin) to **14 days**; airway; daily ECG; notify; vaccinate after recovery.
24	Diphtheria contact management (UCG 2023).	Throat swabs; **benzathine penicillin once** (600,000 IU <10 y; 1.2 MU ≥10 y) or erythromycin 7–10 days; check/complete DTP, booster if last dose >1 year ago; monitor for 7 days.
25	Uganda UNEPI vaccines that prevent URTI complications.	**DTP-HepB-Hib (pentavalent) and PCV at 6, 10, 14 weeks** (diphtheria, Hib epiglottitis, pneumococcal AOM); **measles-rubella at 9 and 18 months** (measles croup/OM). HepB birth dose since 2025.
26	Unilateral foul-smelling nasal discharge in a toddler: diagnosis?	**Nasal foreign body** (bean, groundnut, foam, bead) until proven otherwise. Remove (refer to ENT if not easily removed); treat infection.
27	Why do toddlers get otitis media after a cold?	Their **Eustachian tube is shorter, wider and more horizontal**, and adenoids are large, so nasopharyngeal secretions and bacteria reach the middle ear. Smoke and bottle-feeding lying flat add risk; breastfeeding protects.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 3-year-old girl (14 kg) has had a runny nose, cough and mild fever for 3 days. She is playing, drinking well, RR 28/min, no chest indrawing, SpO_2 98%. The throat is mildly red with no exudate; ear drums are normal. Her mother asks for 'a strong antibiotic so it does not become pneumonia'. What is the MOST appropriate management?

- **A** Amoxicillin 40 mg/kg twice daily for 5 days to prevent pneumonia
- **B** Home care with fluids, saline nose drops and paracetamol, no antibiotic, and advice on danger signs
- **C** Cotrimoxazole for 5 days and a cough syrup
- **D** Benzathine penicillin 600,000 IU IM once
- **E** Admit for observation because she is under 5 years

2. A 10-year-old boy (28 kg) presents with 2 days of sore throat and fever 38.9 °C. He has no cough or runny nose. Both tonsils are enlarged with white exudate and there are tender anterior cervical nodes. There is no rapid strep test. His family lives 20 km away and his mother says he 'never finishes medicine'. What is the BEST treatment?

- **A** Paracetamol only, because most sore throats are viral
- **B** Amoxicillin for 3 days
- **C** Ciprofloxacin 15 mg/kg twice daily for 7 days
- **D** Benzathine penicillin 600,000 IU deep IM as a single dose
- **E** Prednisolone 1 mg/kg for 3 days

3. A 16-month-old boy (10 kg) wakes at 2 a.m. with a barking cough, hoarse cry and noisy breathing after 2 days of a runny nose. In the emergency room he is alert on his mother's lap, has inspiratory stridor at rest with mild subcostal recession, SpO_2 95%, T 37.9 °C. He is drinking. What is the MOST appropriate first treatment?

- **A** Oral dexamethasone 0.15–0.6 mg/kg once
- **B** Examine the throat with a spatula to exclude a foreign body
- **C** IV ceftriaxone 50 mg/kg
- **D** Salbutamol 2.5 mg nebulised every 20 minutes
- **E** Steam inhalation from a kettle for 20 minutes

4. Thirty minutes after oral dexamethasone, the 16-month-old with croup (10 kg) is agitated with loud stridor at rest, marked sternal and subcostal recession and SpO_2 90% in air. What is the MOST appropriate next step?

- **A** A second dose of oral dexamethasone and discharge
- **B** Immediate lateral neck X-ray
- **C** Diazepam 0.3 mg/kg rectally to calm him
- **D** IV ampicillin and gentamicin
- **E** Nebulised adrenaline 1:1000, 4–5 mL, with oxygen, and call senior/anaesthetic help

5. A 4-year-old unimmunised girl (13 kg) from a border district is brought in with high fever for 6 hours. She is sitting upright, leaning forward with her mouth open, drooling, with a soft inspiratory stridor and a muffled voice. She is not coughing. SpO_2 93%. What is the MOST appropriate immediate action?

A Lay her flat and examine the throat with a spatula and torch

B Insert an IV cannula and take blood cultures before anything else

C Keep her with her mother, give oxygen without distressing her, and call the most experienced anaesthetist/ENT to secure the airway in theatre

D Give IM ceftriaxone and send her for a lateral neck X-ray

E Give oral dexamethasone and observe in the waiting area

6. An 18-month-old boy (10 kg) has had fever and has been pulling at his right ear for 2 days after a cold. Otoscopy shows a red, bulging right tympanic membrane with no discharge. He is drinking and has no danger signs. Which treatment is MOST appropriate according to WHO/IMCI practice in Uganda?

A Ciprofloxacin ear drops for 2 weeks

B Wicking only, without antibiotics

C Ceftriaxone 50 mg/kg IV for 5 days

D Amoxicillin 400 mg (40 mg/kg) twice daily for 5 days with paracetamol

E Chlorphenamine and xylometazoline drops for 10 days

7. A 6-year-old HIV-positive boy (17 kg) on ART has had painless foul discharge from the left ear for 6 weeks. He has been given three courses of oral amoxicillin. There is no fever and no swelling behind the ear. Otoscopy shows pus and a central perforation. What is the BEST management?

A A fourth course of oral amoxicillin at a higher dose

B Dry mopping (wicking) three times daily plus ciprofloxacin ear drops for 2 weeks, keep water out, and arrange a hearing assessment

C Admit for IV ceftriaxone and metronidazole

D Insert cotton wool soaked in warm oil and review in a month

E Refer immediately for mastoidectomy

8. A 5-year-old girl (16 kg) with a 2-week history of right ear discharge now has fever 39.2 °C and a tender, fluctuant swelling behind the right ear that pushes the pinna forward and downward. She is alert with no neck stiffness. What is the MOST appropriate management?

A Admit, IV ceftriaxone 80 mg/kg daily plus metronidazole 7.5 mg/kg 8-hourly, and urgent ENT referral for drainage

B Oral amoxicillin and review in 5 days

C Wicking and ciprofloxacin ear drops only

D Incise the swelling in the OPD and send home on paracetamol

E Warm compresses and chlorphenamine

9. A 7-year-old unimmunised boy (20 kg) from a refugee settlement has had a sore throat and low-grade fever for 3 days. He has a markedly swollen neck and a grey-white membrane over both tonsils that bleeds when touched. His voice is hoarse. What is the MOST important immediate treatment?

A Oral amoxicillin and paracetamol

B Remove the membrane with forceps to relieve obstruction

C Nebulised adrenaline and discharge

D Wait for throat culture results before any specific treatment

E Isolate him and give diphtheria antitoxin plus penicillin, with airway monitoring

10. Ten days after admission for diphtheria, a 7-year-old who is now swallowing well suddenly becomes breathless with HR 46/min and a gallop rhythm. What is the MOST likely cause? Advanced

A Aspiration pneumonia
B Anaphylaxis to antitoxin
C Diphtheritic myocarditis with heart block
D Rheumatic carditis
E Hypokalaemia from penicillin

11. A 2-year-old with croup was given dexamethasone and nebulised adrenaline 40 minutes ago. Stridor at rest has disappeared, he is playing and SpO_2 is 97%. The mother wants to go home now. What should you advise?

A Go home now; the adrenaline has cured the croup
B Stay for observation for at least 2–4 hours after adrenaline; admit if stridor at rest returns
C Give a second adrenaline dose before leaving to prevent relapse
D Go home with oral amoxicillin to prevent bacterial tracheitis
E Stay for 24 hours of mandatory IV fluids

12. A 15-year-old girl (48 kg) has fever, severe sore throat with exudative tonsillitis, posterior cervical lymphadenopathy, marked fatigue and a palpable spleen tip. Which drug should you AVOID? Advanced

A Paracetamol
B Ibuprofen with food
C Salt-water gargles
D Penicillin V if GAS is confirmed
E Amoxicillin

13. A 9-year-old boy (25 kg) treated for tonsillitis with 3 days of amoxicillin now has severe left-sided throat pain, cannot open his mouth fully, is drooling, and speaks with a muffled voice. The left soft palate bulges and the uvula is deviated to the right. What is the MOST likely diagnosis and next step?

A Peritonsillar abscess: admit, IV penicillin plus metronidazole, and ENT drainage
B Epiglottitis: secure the airway in theatre before anything else
C Diphtheria: give antitoxin
D Viral pharyngitis: reassure and continue amoxicillin
E Dental abscess: extract the tooth in OPD

14. A 6-year-old girl has had a blocked nose with purulent discharge and daytime cough for 14 days without any improvement. She is afebrile with mild tenderness over the maxillary sinuses. There is no eye swelling. Which is the MOST appropriate treatment?

A No treatment; sinusitis is always viral
B Chlorphenamine and xylometazoline for 4 weeks
C IV ceftriaxone and urgent CT of the sinuses
D Amoxicillin for 7–10 days with saline irrigation and analgesia
E Ciprofloxacin orally for 14 days

15. A 3-year-old with croup has been in hospital for 2 days and was improving. Today he has a temperature of 39.8 °C, looks toxic, and has worsening stridor with thick secretions; two doses of nebulised adrenaline give little improvement. What is the MOST likely diagnosis? Advanced

A Rebound after adrenaline
B Epiglottitis
C Bacterial tracheitis
D Retropharyngeal abscess
E Laryngomalacia

Answers & explanations

1. B · Home care with fluids, saline nose drops and paracetamol, no antibiotic, and advice on danger signs

B is right: this is IMCI **'no pneumonia: cough or cold'**, a viral URTI; the treatment is home care, symptom relief and a clear safety net. A is wrong: antibiotics do not prevent pneumonia after a cold and cause harm. C is wrong: no antibiotic is indicated and UCG 2023 advises against cough syrups in children <6 years. D is wrong: there are no features of streptococcal tonsillitis (and GAS is uncommon at 3 years). E is wrong: she has no danger signs or distress; admission is unnecessary.

2. D · Benzathine penicillin 600,000 IU deep IM as a single dose

D is right: McIsaac score 5 (fever, exudate, tender nodes, no cough, age 3–14) → likely **GAS**. With doubtful adherence, a **single benzathine penicillin injection** (600,000 IU if <30 kg) guarantees eradication and prevents rheumatic fever. A is wrong: this is a high-probability bacterial tonsillitis in a high-RHD country. B is wrong: 3 days does not eradicate GAS; a 10-day course is needed. C is wrong: ciprofloxacin has poor GAS activity and is not indicated. E is wrong: steroids may relieve pain but do not treat infection or prevent rheumatic fever.

3. A · Oral dexamethasone 0.15–0.6 mg/kg once

A is right: this is **moderate croup** (stridor at rest, recession, alert). Every child with croup and stridor should get a single dose of **oral dexamethasone** (0.15 mg/kg UCG; up to 0.6 mg/kg). B is wrong: throat examination can upset the child and worsen obstruction, and the history (prodrome, barking cough) points to croup, not a foreign body. C is wrong: croup is viral; antibiotics are not indicated. D is wrong: salbutamol treats lower-airway bronchospasm, not subglottic oedema. E is wrong: humidified air has no proven benefit and steam kettles cause scalds.

4. E · Nebulised adrenaline 1:1000, 4–5 mL, with oxygen, and call senior/anaesthetic help

E is right: he now has **severe croup** with hypoxaemia. Give **nebulised adrenaline 1:1000 at 0.4–0.5 mL/kg (max 5 mL)** with oxygen, alert the airway team, and observe for at least 2–4 hours. A is wrong: steroids take time to work; discharging a hypoxic child is dangerous. B is wrong: X-rays delay treatment and move a child with a critical airway away from help. C is wrong: sedation can precipitate complete obstruction and respiratory arrest. D is wrong: there is no evidence of bacterial infection.

5. C · Keep her with her mother, give oxygen without distressing her, and call the most experienced anaesthetist/ENT to secure the airway in theatre

C is right: this is **epiglottitis** (toxic, drooling, tripod, muffled voice, no cough, unvaccinated). The priority is a calm child and a **secured airway** by the most skilled person, in theatre. A is wrong: lying flat and a spatula can cause complete obstruction. B is wrong: painful procedures before the airway is secure can precipitate obstruction (UCG: avoid procedures that agitate the child). D is wrong: IM injections should be avoided, and sending her to X-ray removes her from airway help. E is wrong: this is not croup; observation in a waiting area could be fatal.

6. D · Amoxicillin 400 mg (40 mg/kg) twice daily for 5 days with paracetamol

D is right: **acute otitis media** in a child <2 years; WHO pocket book/IMCI: **amoxicillin 40 mg/kg twice daily for 5 days** plus analgesia, review in 5 days. A is wrong: topical quinolones are for CSOM (discharge ≥14 days), and the drum is intact. B is wrong: wicking is for a discharging ear; this child needs an antibiotic and analgesia. C is wrong: IV ceftriaxone is for complications or treatment failure. E is wrong: antihistamines and decongestants do not treat AOM.

7. B · Dry mopping (wicking) three times daily plus ciprofloxacin ear drops for 2 weeks, keep water out, and arrange a hearing assessment

B is right: discharge ≥14 days through a perforation = **CSOM**: wicking + **topical quinolone** drops for 2 weeks; systemic antibiotics are not useful (UCG 2023); assess hearing and refer to ENT if it persists. A is wrong: repeated oral antibiotics fail in CSOM and breed resistance. C is wrong: there are no signs of complications (mastoiditis, meningitis) to justify IV therapy. D is wrong: oils and plugs keep the ear wet and infected. E is wrong: there is no mastoid swelling or tenderness; surgery is not the first step.

8. A · Admit, IV ceftriaxone 80 mg/kg daily plus metronidazole 7.5 mg/kg 8-hourly, and urgent ENT referral for drainage

A is right: this is **mastoiditis** with probable subperiosteal abscess: admit, **ceftriaxone 50–80 mg/kg + metronidazole** for 10–14 days (UCG 2023), and urgent ENT for drainage/mastoidectomy, watching for intracranial spread. B is wrong: oral treatment is inadequate for a bone infection with abscess. C is wrong: topical therapy cannot treat mastoid infection. D is wrong: drainage must be done by ENT under appropriate conditions with IV antibiotics; OPD incision risks facial nerve injury and incomplete drainage. E is wrong: these do nothing for a bacterial abscess.

9. E · Isolate him and give diphtheria antitoxin plus penicillin, with airway monitoring

E is right: an **unimmunised** child with an adherent grey membrane that bleeds and a **bull neck** has **diphtheria**. Antitoxin must be given **on clinical suspicion** (it neutralises only unbound toxin), with penicillin (or a macrolide) for 14 days, isolation, airway care, ECG monitoring and contact management. A is wrong: amoxicillin alone does not neutralise toxin. B is wrong: stripping the membrane causes bleeding and can worsen obstruction or toxin absorption. C is wrong: adrenaline does not treat diphtheria and discharge would be dangerous. D is wrong: waiting for culture loses the window for antitoxin.

10. C · Diphtheritic myocarditis with heart block

C is right: diphtheria toxin causes **myocarditis typically in days 7–14**, with heart block, arrhythmias and heart failure; daily ECG and strict bed rest are needed, with cardiac support. A is wrong: aspiration causes fever and focal chest signs, not bradycardia and gallop. B is wrong: anaphylaxis to antitoxin occurs within minutes to hours of the infusion, not 10 days later (serum sickness occurs at 7–10 days but causes rash, fever and arthralgia, not heart block). D is wrong: rheumatic fever follows GAS, not diphtheria. E is wrong: penicillin does not cause hypokalaemia of this kind.

11. B · Stay for observation for at least 2–4 hours after adrenaline; admit if stridor at rest returns

B is right: nebulised adrenaline works for about 2 hours and symptoms can **rebound**; observe at least 2–4 hours and admit if stridor at rest recurs. A is wrong: adrenaline is temporary; the steroid takes longer to act. C is wrong: adrenaline is given for current severe obstruction, not as prophylaxis. D is wrong: antibiotics don't prevent tracheitis and croup is viral. E is wrong: he is drinking; IV fluids are unnecessary.

12. E · Amoxicillin

E is right: this picture (teenager, exudative tonsillitis, **posterior** nodes, **splenomegaly**, fatigue) suggests **infectious mononucleosis (EBV)**; **amoxicillin** commonly causes a widespread maculopapular rash in EBV infection. A is wrong: paracetamol is safe symptomatic treatment. B is wrong: ibuprofen with food is reasonable for pain. C is wrong: gargles are harmless symptomatic care. D is wrong: penicillin V can be used if co-existing GAS is proven; it is far less likely than amoxicillin to cause the EBV rash. (She should also avoid contact sports while the spleen is enlarged.)

13. A · Peritonsillar abscess: admit, IV penicillin plus metronidazole, and ENT drainage

A is right: unilateral pain, **trismus**, drooling, muffled voice and a **bulging palate with uvula pushed to the opposite side** = **quinsy**; needs IV antibiotics (penicillin + metronidazole) and **needle aspiration or incision** by ENT. B is wrong: epiglottitis is a midline supraglottic infection without trismus or palatal bulge. C is wrong: there is no membrane and he is presumably immunised. D is wrong: these are not features of a simple viral sore throat. E is wrong: the findings are in the tonsillar fossa, not a tooth.

14. D · Amoxicillin for 7–10 days with saline irrigation and analgesia

D is right: symptoms **>10 days without improvement** meet criteria for **acute bacterial sinusitis**: amoxicillin for 7–10 days (UCG 2023), saline irrigation and analgesia. A is wrong: persistent symptoms beyond 10 days suggest a bacterial infection that benefits from antibiotics. B is wrong: decongestant drops should not be used beyond 5 days (rebound) and antihistamines don't treat infection. C is wrong: IV therapy and CT are for orbital or intracranial complications. E is wrong: ciprofloxacin is not first line for pneumococcus/H. influenzae sinusitis in children.

15. C · Bacterial tracheitis

C is right: croup that **suddenly worsens** with **high fever, toxicity, purulent secretions** and **poor response to adrenaline** suggests **bacterial tracheitis** (S. aureus, GAS): secure the airway and give IV antibiotics. A is wrong: rebound occurs within hours of adrenaline and does not cause high fever or toxicity. B is wrong: epiglottitis begins abruptly, not after days of croup, and children have little cough. D is wrong: retropharyngeal abscess causes neck stiffness, drooling and torticollis, typically without a croup prodrome. E is wrong: laryngomalacia is chronic stridor from early infancy, not an acute febrile illness.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Night-time stridor at Mengo emergency

Daniel, a 14-month-old boy, is brought at 1 a.m. with a 'cough like a dog' and noisy breathing that began tonight after 2 days of a runny nose and mild fever. He is on his mother's lap, crying intermittently. His immunisation card shows BCG, OPV, pentavalent ×3, PCV ×3 and MR1 at 9 months.

1. What is your immediate assessment, and what will you avoid?

> ETAT: look for **stridor at rest, recession, cyanosis, SpO_2, consciousness, drooling**. Keep him **on his mother's lap**, examine gently, **avoid** throat examination with a spatula, lying him flat, painful procedures and separation from the mother.

2. Findings: weight 9 kg (WAZ −0.9), T 38.0 °C, RR 40, HR 145, SpO_2 94%, inspiratory stridor at rest with moderate subcostal recession, alert, no drooling, drinking. Classify.

> **Moderate croup** (stridor at rest, recession, alert; Westley about 4). Probable viral (parainfluenza). Not epiglottitis (no toxicity, drooling or tripod; he has a barking cough and is drinking).

3. Write the initial treatment.

> **Dexamethasone 0.15–0.6 mg/kg orally once** (1.4–5.4 mg; UCG 0.15 mg/kg; many units give 0.6 mg/kg) or prednisolone 1–2 mg/kg; paracetamol 135 mg (15 mg/kg); keep calm; observe SpO_2 and work of breathing every 15–30 min. No antibiotics, no steam, no cough syrup.

4. Forty minutes later he is agitated, stridor is louder, recession marked, SpO_2 89%. What now?

> **Severe croup**: **nebulised adrenaline 1:1000, 0.5 mL/kg = 4.5 mL** (max 5 mL), oxygen held near the face, call the senior and anaesthetist; repeat adrenaline after 30 min if needed; prepare for intubation with a tube 0.5–1 mm smaller than usual if he tires.

5. He improves after one adrenaline nebulisation (no stridor at rest, SpO_2 96%). Disposition?

> **Admit** (needed adrenaline, age 14 months, hypoxaemia at presentation, night time). Observe for **rebound** for at least 2–4 h; continue oral fluids; repeat steroid at 12–24 h if stridor persists; oxygen if SpO_2 <92%.

6. On day 2 the mother asks whether he needs 'the measles injection' and antibiotics. How do you respond?

> Croup is viral; **antibiotics are not needed** unless signs of bacterial tracheitis or pneumonia appear. Check for measles features (rash, conjunctivitis, Koplik spots); his **MR2 is due at 18 months** (UNEPI). Discharge when no stridor at rest and drinking, with advice to return for stridor at rest, recession, drooling or blue colour.

Take-away: Croup is graded by stridor at rest, recession, SpO_2 and consciousness: every child gets a steroid, severe croup gets nebulised adrenaline with oxygen, and anyone given adrenaline is observed for rebound.

Case 2: A sore throat and a heart at risk

Grace, an 11-year-old girl from Mbale, comes to the HC IV with 2 days of sore throat, fever and painful swallowing. Her older brother has rheumatic heart disease and receives monthly injections at the regional referral hospital.

1. What features in the history and examination would point to streptococcal tonsillitis rather than a viral sore throat?

> Age 3–15, **fever >38 °C**, **tonsillar exudate**, **tender anterior cervical nodes**, **no cough or coryza**, palatal petechiae, scarlet rash. Viral clues: cough, runny nose, hoarseness, conjunctivitis, ulcers, diarrhoea.

2. She is 32 kg, T 38.7 °C, exudative tonsils, tender anterior nodes, no cough, no murmur. Score her and decide.

> **McIsaac 5** → high probability of GAS. In Uganda (high RHD burden, family history of RHD, no rapid test): **treat**.

3. Prescribe treatment, with dose and duration.

> Either **benzathine penicillin 1.2 MU deep IM once** (≥30 kg), or **penicillin V 500 mg (max dose) 6-hourly for 10 days** (UCG 10–20 mg/kg/dose). Penicillin allergy: erythromycin 12.5 mg/kg (400 mg) 6-hourly × 10 days. Plus paracetamol 15 mg/kg (480 mg) 6-hourly and fluids. Observe 30 min after the injection with adrenaline ready.

4. Why does the duration matter, and what do you tell the family?

> A full **10-day course (or the injection)** eradicates GAS and **prevents acute rheumatic fever**; stopping when the pain settles (day 2–3) leaves bacteria behind. Tell them to **return if joint pains, breathlessness, chest pain, jerky movements or dark urine/puffy face** appear in the next weeks.

5. Three days later she cannot open her mouth fully, is drooling, and her voice is muffled; the right palate bulges. What has happened and what do you do?

> **Right peritonsillar abscess (quinsy)**. Admit, IV fluids, **IV benzylpenicillin 50,000 IU/kg 6-hourly + metronidazole 7.5 mg/kg 8-hourly**, analgesia, and **refer to ENT for aspiration/incision and drainage**; complete 10–14 days of antibiotics.

> **Take-away:** In Uganda, a high-scoring sore throat in a school-age child is treated with a full 10 days of penicillin or a single benzathine injection, because what you are really preventing is rheumatic heart disease.

Case 3: The drooling child from the border

Amina, a 5-year-old girl, arrives from a refugee settlement in Isingiro with high fever since this morning. She is sitting on her father's lap, leaning forward with her chin out and mouth open, drooling. She has never been vaccinated.

1. What are your top two diagnoses and what single action protects her most?

> **Epiglottitis** (unvaccinated, toxic, drooling, tripod) and **diphtheria** (unvaccinated; membrane, bull neck); also retropharyngeal abscess and bacterial tracheitis. The single most protective action: **keep her calm and upright with her father, oxygen without distress, and call the most experienced anaesthetist/ENT now**.

2. What must you not do?

> Do **not** lie her down, examine the throat with a spatula, insert cannulae or give IM injections, or send her alone for an X-ray before the airway is secured (UCG 2023).

3. In theatre the anaesthetist sees a cherry-red, swollen epiglottis and intubates her. What treatment follows?

> IV access, blood culture, **ceftriaxone 50–100 mg/kg IV once daily for 7–10 days** (16 kg → 800 mg–1.6 g; UCG 50 mg/kg), ICU/HDU care until the swelling settles (usually 24–48 h), sedation while intubated, then extubation in a controlled setting.

4. Her 2-year-old brother, also unvaccinated, lives in the same tent. What do you do for the family?

> **Rifampicin prophylaxis 20 mg/kg (max 600 mg) once daily for 4 days** for household contacts (an unvaccinated child <4 y is present); **start catch-up vaccination** (pentavalent with Hib, PCV, MR, OPV/IPV) for both children; notify the settlement health team.

5. How would the picture differ if this were diphtheria, and what would you add?

> Diphtheria: **slower onset**, low-grade fever, **grey adherent membrane** that bleeds when removed, **bull neck**, hoarseness. Add **isolation**, **diphtheria antitoxin** (20,000–100,000 IU by severity) on suspicion, penicillin or erythromycin for 14 days, **daily ECG** (myocarditis days 7–14), contact prophylaxis, notification, and vaccination.

Take-away: A drooling, toxic, unvaccinated child needs a calm parent, oxygen and the best airway person, not a spatula; then antibiotics, contact prophylaxis and the vaccines she missed.

References

1. Uganda Clinical Guidelines 2023 (5.2.3 Coryza; 5.2.4 Epiglottitis; 5.2.7 Croup; 21.1.4 Otitis media; 21.1.6 Mastoiditis; 21.2.4 Sinusitis; 21.3.3–21.3.5 Pharyngitis, tonsillitis, quinsy; 2.1.3 Diphtheria). *Ministry of Health Uganda*, 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
2. Pocket Book of Hospital Care for Children, 2nd edition. *World Health Organization*, 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
3. Sore throat (acute): antimicrobial prescribing (NG84). *National Institute for Health and Care Excellence*, 2018. https://www.nice.org.uk/guidance/ng84
4. Croup (laryngotracheobronchitis): Clinical Practice Guideline. *The Royal Children's Hospital Melbourne*. https://www.rch.org.au/clinicalguide/guideline_index/Croup_laryngotracheobronchitis/
5. Croup. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/962972-overview
6. Epiglottitis. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/963773-overview
7. Pediatric Pharyngitis. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/967384-overview
8. Otitis Media. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/994656-overview
9. Diphtheria: fact sheet. *World Health Organization*. https://www.who.int/news-room/fact-sheets/detail/diphtheria

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.